

Internet of Things

Department of Computer Security and Network Systems

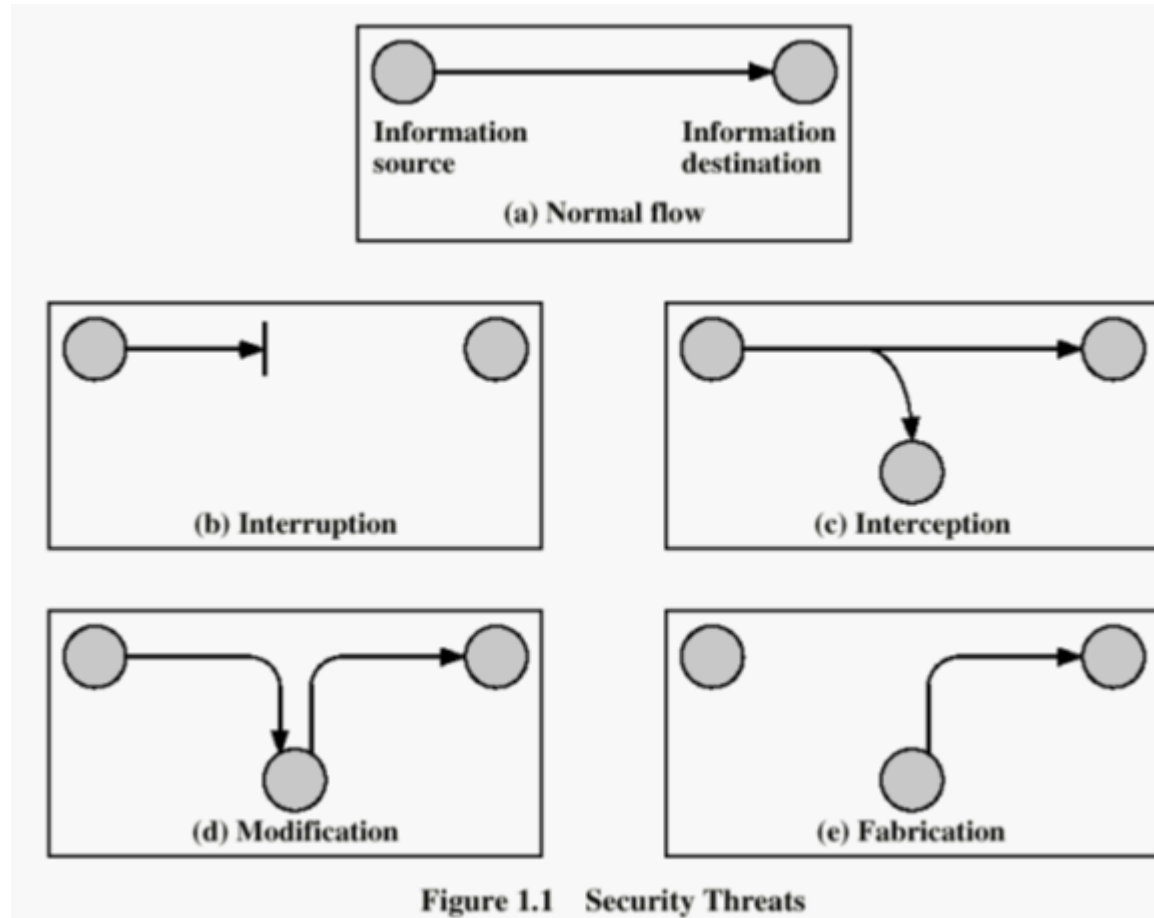
IOT THREATS



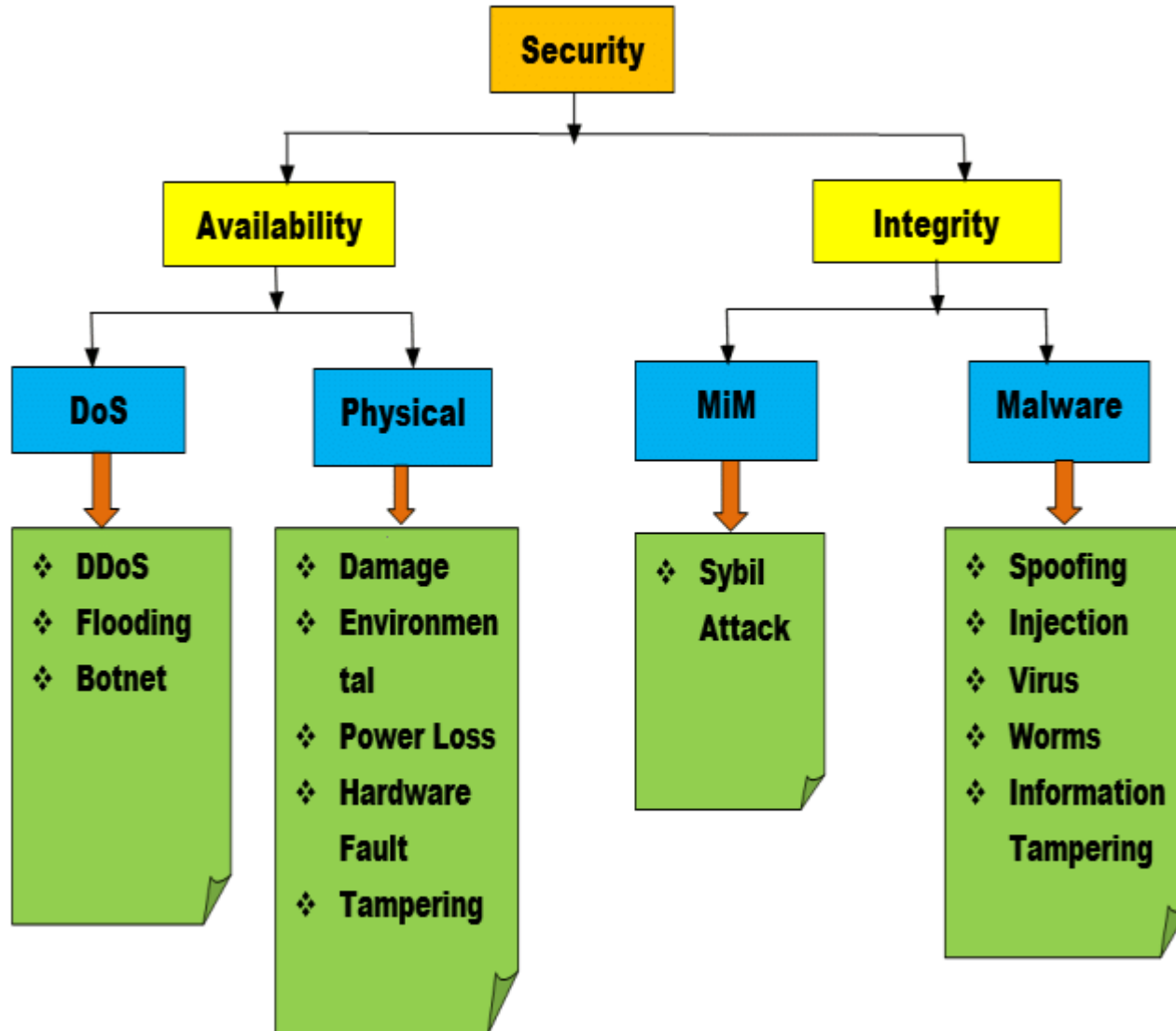
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Threats



Threats



Threat Types

Internal Threats

External Threats



Types of Cybersecurity Threats



1.Malware: Viruses, worms, Trojans, ransomware

1.Phishing: Fraudulent attempts to obtain sensitive data

2.Denial of Service (DoS) and Distributed Denial of Service (DDoS): Overloading systems to disrupt operations.

3.Man-in-the-Middle (MitM) Attacks: Intercepting communications

4.Insider Threats: Malicious or negligent actions by employees

5.Advanced Persistent Threats (APTs): Long-term, targeted attacks

6.Zero-Day Exploits: Attacking vulnerabilities before they are patched

Real-World Examples of Attacks

- Case 1:** Wannacry Ransomware Attack (2017)

Impact: Affected 200,000+ computers in 150 countries

- Case 2:** SolarWinds Supply Chain Attack (2020)

Impact: Targeted US federal agencies and large enterprises

- Case 3:** Colonial Pipeline Ransomware Attack (2021)

Impact: Disrupted fuel supply, leading to shortage



Real-World Examples of Attacks

1. Mozi Botnet Activity

- **What It Is:** Mozi is a peer-to-peer (P2P) botnet predominantly targeting IoT devices such as routers and DVRs.
- **Key Characteristics:**
 - Uses a distributed hash table (DHT) for communication, making it harder to shut down.
 - Spreads by brute-forcing weak/default credentials and by exploiting known vulnerabilities in IoT devices.
- **Recent Developments:**
 - Since its emergence in late 2019, Mozi has continued to evolve. Security researchers reported ongoing campaigns throughout 2022 and 2023.
 - It is often used for large-scale Distributed Denial of Service (DDoS) attacks and can also deploy other malicious payloads.



Real-World Examples of Attacks



2. Mirai Variants (e.g., Satori, Mozi-Influenced Strains, etc.)

- **What It Is:** Mirai is one of the most well-known IoT botnets, first discovered in 2016. Attackers have repeatedly modified its source code—released publicly in 2016—leading to new variants that persist to this day.
- **Recent Developments:**
 - Satori (also called Mirai Okiru) continues to pop up in 2022–2023 with new exploits targeting various routers and IoT devices.
 - Some variants incorporate exploits for recent router vulnerabilities (e.g., vulnerabilities in Realtek SDK, GPON home routers, or Huawei devices).
 - Attackers use these botnets to launch DDoS attacks, cryptomining, or data exfiltration.

3. RapperBot Campaign

- **What It Is:** RapperBot, discovered in mid-2022, is a brute-forcing botnet targeting Linux-based IoT devices.
- **Key Characteristics:**
 - Focuses on SSH brute-force attacks rather than the more common Telnet approach used by many older botnets.
 - Can install its own SSH backdoor, enabling persistent remote access.
- **Recent Developments:**
 - Continues to evolve, with the operators frequently updating command-and-control (C2) infrastructure and infection methods.
 - Mostly used for DDoS attacks, but its ability to maintain persistent access is concerning for potential data theft or further malware deployment.

Real-World Examples of Attacks



4. Zerobot (Golang-Based Botnet)

- **What It Is:** Zerobot, first publicly discussed in late 2022, is a botnet written in Go (Golang) with modular capabilities.
- **Key Characteristics:**
 - Exploits known vulnerabilities in IoT and web applications (e.g., in F5 BIG-IP, D-Link devices, vulnerable Apache servers).
 - Written in a cross-platform language (Go), making it easier to port to different device architectures.
- **Recent Developments:**
 - Continues to add new exploits and functionalities (crypto-mining modules, self-propagation).

Often used in DDoS for hire (“booter” or “stresser” services).

5. Chaos / Kaiji Botnets

- **What It Is:** Chaos (a rebrand or evolution of Kaiji) is a multi-platform malware family that targets both Linux-based systems and Windows. When targeting IoT devices, it attempts to brute force SSH/Telnet credentials.
- **Key Characteristics:**
 - DDoS capabilities, cryptomining, and the ability to self-propagate by exploiting IoT vulnerabilities.
 - Some variations focus on cloud environments (e.g., AWS, Azure) when they gain a foothold in larger networks.
- **Recent Developments:**
 - Seen in 2022–2023 waves, especially exploiting weak credentials on routers and network-attached storage (NAS).
 - Highly flexible: the malware authors regularly update modules to adapt to new exploits.

Real-World Examples of Attacks



6. Ongoing Exploitation of Home Routers & Cameras

- **Trend Overview:** Beyond named botnets, many attacks revolve around systematically scanning the internet for exposed IoT devices—particularly home routers, cameras, or network-attached storage devices.
- **Tactics:**
 - Attackers exploit default passwords or unpatched vulnerabilities.
 - Once compromised, devices can be enrolled into botnets or used as entry points to larger corporate/home networks.
- **Recent Examples:**
 - Vulnerabilities in certain Realtek-based routers (CVE-2021-35395 and others) continued to be exploited into 2023.
 - IP cameras with outdated firmware remain a prime target, as they often have remote viewing enabled and are seldom updated.

7. Wiper-Style Attacks (e.g., Silex)

- **What It Is:** Although less common than DDoS-oriented botnets, some malware (like Silex in 2019) focused on “bricking” devices by wiping their storage or corrupting firmware.
- **Recent Developments:**
 - While Silex itself tapered off, security researchers occasionally see “wiper” functionalities in new IoT threats, hinting that destructive attacks remain a possibility.
 - Such attacks can render devices unusable and require manual re-flashing or even complete replacement.

Real-World Examples of Attacks

Microsoft Exchange Remote Code Execution Attack

In March 2021, a large-scale cyber attack was carried out against Microsoft Exchange email server. It leveraged four separate zero-day vulnerabilities discovered in Microso



These vulnerabilities enable attackers to forge untrusted URLs, use them to access an Exchange Server system, and provide a direct server-side storage path for malware. It is a Remote Code Execution (RCE) attack, which allows attackers to completely compromise a server and gain access to all its data. On affected servers, attackers stole sensitive information, injected ransomware, and deployed backdoors in a way that was almost untraceable.

In the United States alone, the attacks affected nine government agencies and more than 60,000 private businesses.

Real-World Examples of Attacks

Twitter Celebrities Attack

In July 2020, Twitter was breached by a group of three attackers, who took over popular accounts. They used social engineering attacks to steal employee credentials and gain access to internal management systems, later identified by Twitter as vishing (phone phishing).

Dozens of well-known accounts were hacked, including Barack Obama, Jeff Bezos, and Elon Musk. The attackers used the stolen accounts to post bitcoin scams and earned more than \$100,000. Two weeks after the events, the US Justice Department charged three suspects, one of whom was 17 years old at the time.



Why IoT Devices Remain Vulnerable

1. **Default or Weak Credentials:** Many IoT products still ship with easily guessable usernames and passwords (e.g., “admin/admin”) and are never changed by the end-user.
2. **Lack of Updates/Patches:** Manufacturers may not provide regular firmware updates, or end-users fail to apply them.
3. **Resource Constraints:** IoT devices often have minimal memory/processing power, limiting built-in security features.
4. **Constant Internet Exposure:** Many IoT devices are accessible via the public internet (sometimes inadvertently), making them easy targets for scanning and exploitation.



Emerging Cybersecurity Threats

- **AI-Powered Cyberattacks:** Automated phishing and deepfake scams
- **IoT Vulnerabilities:** Exploiting connected devices
- **Cloud Security Risks:** Misconfigurations and data breaches
- **Supply Chain Attacks:** Targeting third-party vendors
- **Quantum Computing Risks:** Breaking traditional encryption



Impact of Cybersecurity Threats

- **Data Breaches:** Exposure of personal and business data
- **Financial Losses:** Cybercrime costs billions annually
- **Operational Disruptions:** Downtime and loss of productivity
- **Reputation Damage:** Loss of customer trust and brand value
- **Legal Consequences:** Fines for failing to protect data



IoT Threats

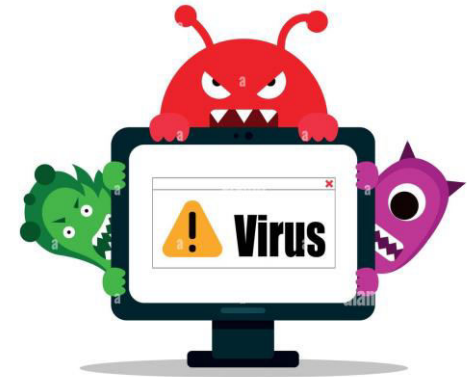
- IoT Growth and Connectivity: The number of IoT devices worldwide is projected to reach 30.9 billion by 2025, creating a massive interconnected network.
- Increased Vulnerabilities: IoT devices often lack robust security features, making them prime targets for cyberattacks.
- Global Importance of IoT Security: A 2023 report revealed that 43% of organizations experienced IoT-related breaches, emphasizing the need for proactive security measures.

Virus

A virus has similarities to biological viruses. Once find a host, it will replicate and distribute them attached to a file.

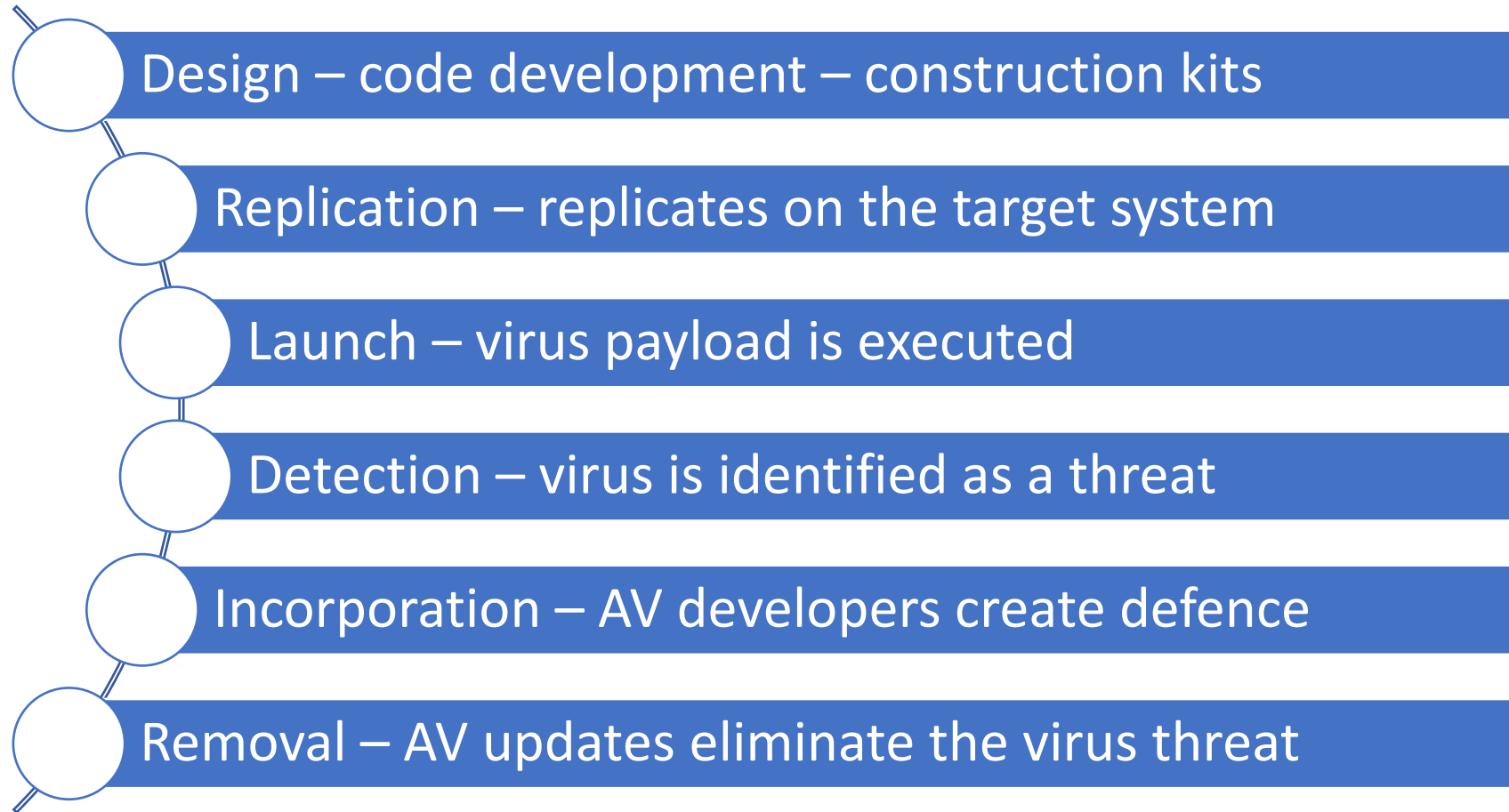
Virus classifications,

1. Boot Sector Virus
2. File Deleting Viruses
3. Mass Mailer Viruses
4. Macro viruses
5. Polymorphic Viruses
6. Armored Viruses
7. Stealth viruses
8. Retrovirus

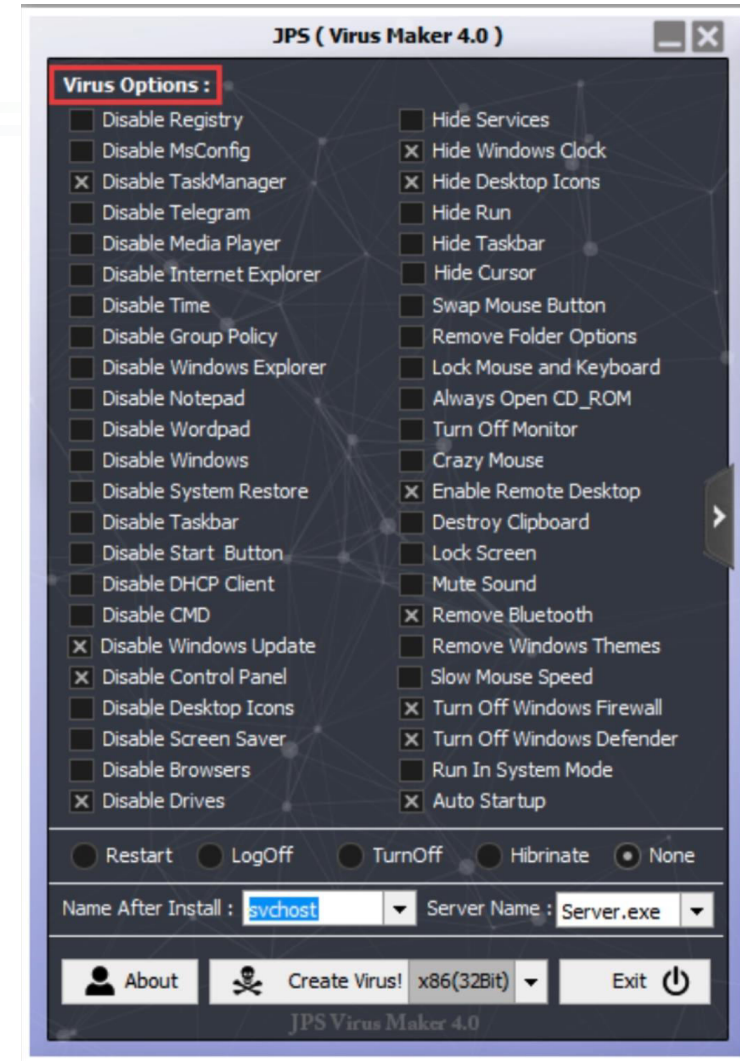
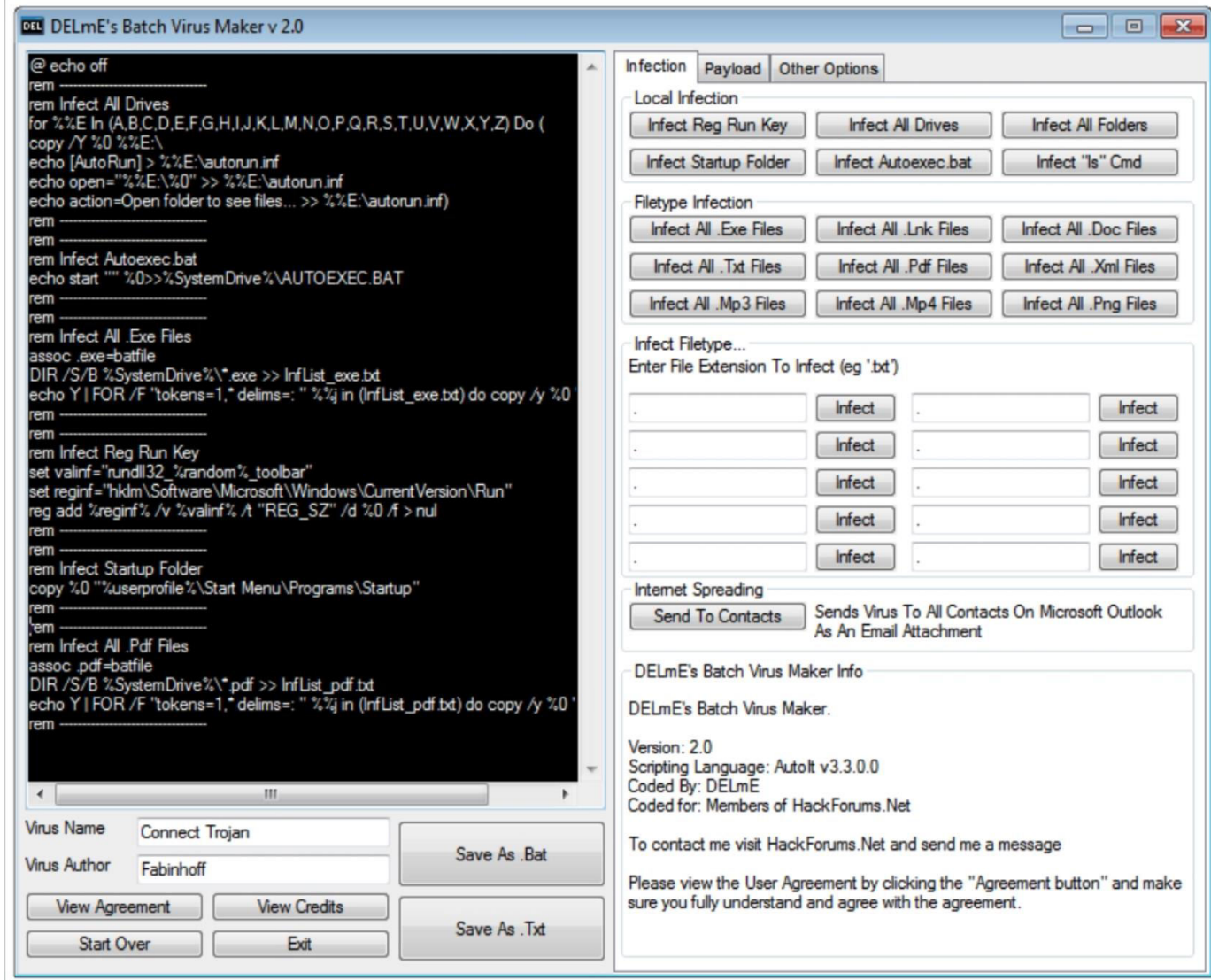


ILOVEYOU virus, 2000: This malware virus impacted millions of computers around the globe and was downloaded by clicking on an attachment called “LOVE-LETTER-FOR-YOU.TXT.vbs” and from an email with the subject line “ILOVEYOU.”

Stages of Virus Lifecycle



Virus Maker Tool



Virus

1. Boot Sector Virus

Virus infects the first sector of the hard drive, where the Master Boot Record (MBR) is stored. The Master Boot Record (MBR) stores the disk's primary partition

Examples :- Elk Cloner, Brain, Stoned, Parity Boot Virus, Michelangelo, and Angelina



Virus

2. File Deleting Viruses

designed to delete critical files which are the part of Operating System or data files like CIH viruses (also known as the Chernobyl virus) it's threatened to delete files automatically and erase the core system code kept in flash memory on the motherboard of certain types of computers

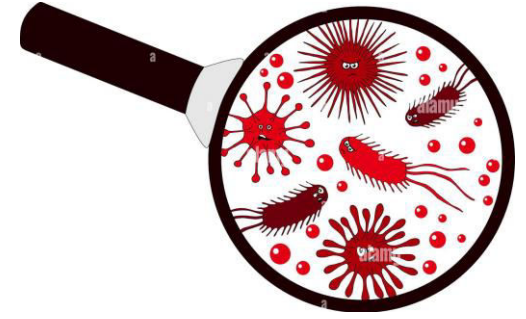




3. Mass Mailer Viruses

Mass Mailer Viruses search e-mail programs like MS outlook for e-mail addresses which are stored in the address book and replicate by e-mailing themselves to the addresses stored in the address book of the e-mail program.

ILOVEYOU virus, 2000: This malware virus impacted millions of computers around the globe and was downloaded by clicking on an attachment called “LOVE-LETTER-FOR-YOU.TXT.vbs” and from an email with the subject line “ILOVEYOU.”



4. Macro viruses

Macro viruses are written by using the Macro programming languages like VBA, which is a feature of MS office package. A macro is a way to automate and simplify a task that you perform repeatedly in MS office suit (MS Excel, MS word etc). These macros are usually stored as part of the document or spreadsheet and can travel to other systems when these files are transferred to another computers.

Example **Melissa Virus**

Virus

5. Polymorphic Viruses

It has the capability to change their appearance and change their code every time they infect a different system. This helps the Polymorphic Viruses to hide from anti-virus software.



Example **Ursnif (also known as Gozi)**



6. Armored Viruses:

This type of viruses that are designed and written to make itself difficult to detect or analyze. An Armored Virus may also have the ability to protect itself from antivirus programs, making it more difficult to disinfect.

Example **Whale virus**



7. Stealth viruses:

Stealth viruses have the capability to hide from operating system or anti-virus software by making changes to file sizes or directory structure. Stealth viruses are anti-heuristic nature which helps them to hide from heuristic detection.

Example **Brain, the very first DOS virus.**

Virus

8. Retro virus:

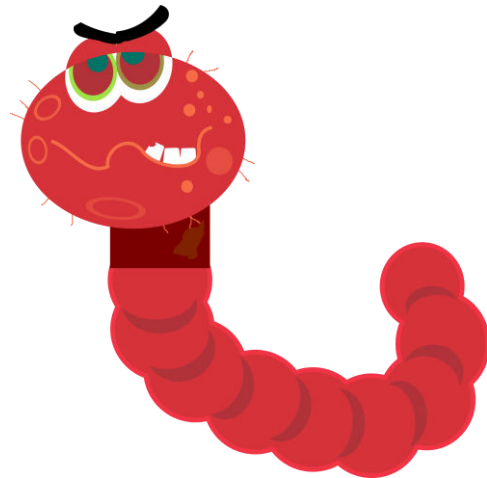
Retrovirus is another type virus which tries to attack and disable the anti-virus application running on the computer. A retrovirus can be considered anti-antivirus. Some Retroviruses attack the anti-virus application and stop it from running or some other destroys the virus definition database.



Worms

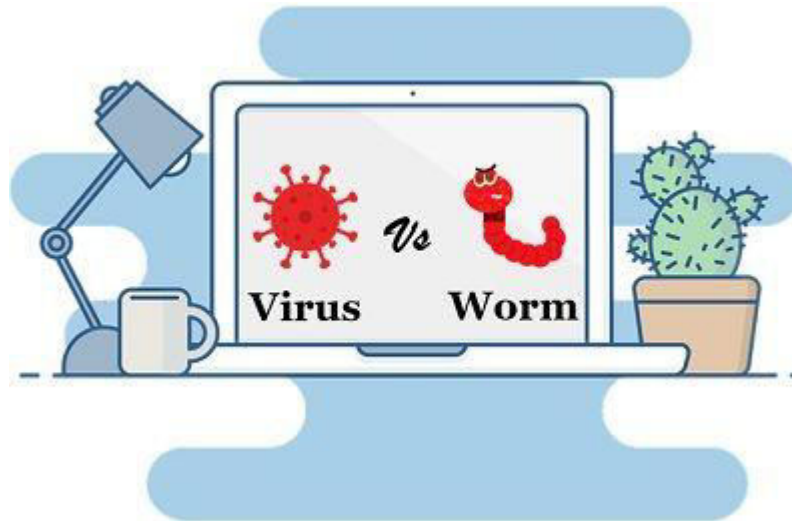
"self-replicating programs that spread with no human intervention after they are started."

As defined in the "Security of the Internet" report released in 1996 by the CERT Division of the Software Engineering Institute at Carnegie Mellon University



Virus vs Worms

"self-replicating programs, but usually require some action on the part of the user to spread inadvertently to other programs or systems."



Worms

- Email worms
- File-sharing worms
- Crypto worms
- Internet worms
- Instant messaging worms



Email worms

Email worms work by creating and sending outbound messages to all the addresses in a user's contact list. The messages include a malicious executable file that infects the new system when the recipient opens it.



File-sharing worms

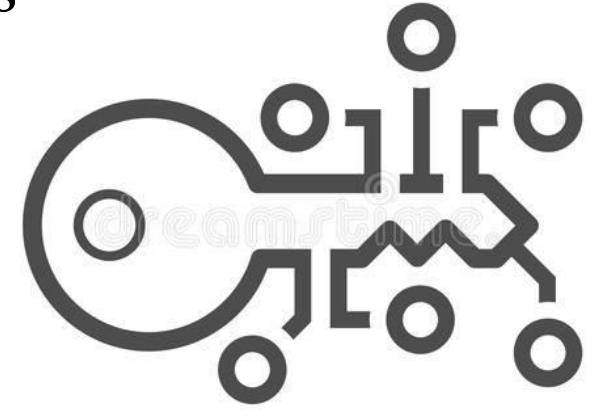
File-sharing worms copy themselves into shared folders and spread through peer-to-peer file-sharing networks.

Worm authors often disguise these malicious programs as media files.



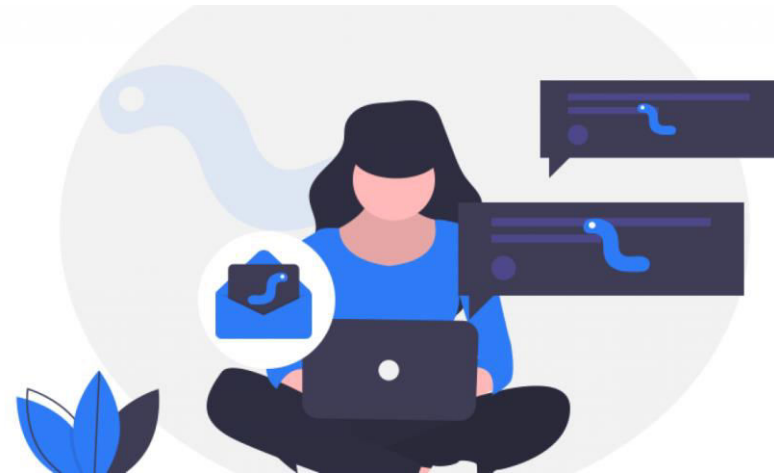
Crypto worms

Crypto worms work by encrypting data on the victim's system. Perpetrators can use this type of worm in ransomware attacks, where they follow up with the victim and demand payment in exchange for a key to decrypt the files.



Instant messaging worms

Like email worms, instant messaging worms are masked by attachments or links, which the worm continues to spread to the infected user's contact list. The only difference is that instead of arriving in an email, it comes as an instant message on a chat service.



Crypto worms

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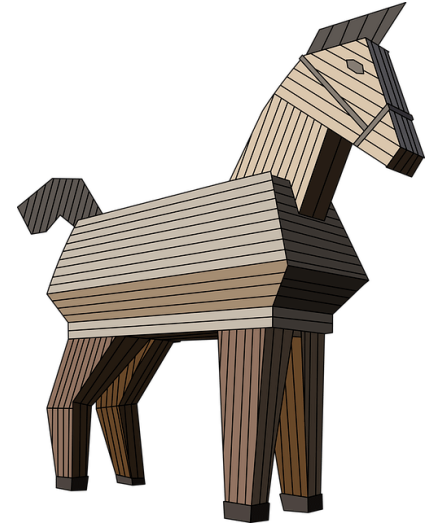


Trojan Horse

A Trojan Horse Virus is a type of malware that downloads onto a computer disguised as a legitimate program.

Actions include:

- Deleting data
- Blocking data
- Modifying data
- Copying data
- Disrupting the performance of computers or computer networks



Trojan Horse

Type of Trojan horse

- Backdoor Trojans
- DDoS Trojans
- Data Sending Trojans
- Destructive Trojans
- Proxy Trojans
- Security software disabler Trojans

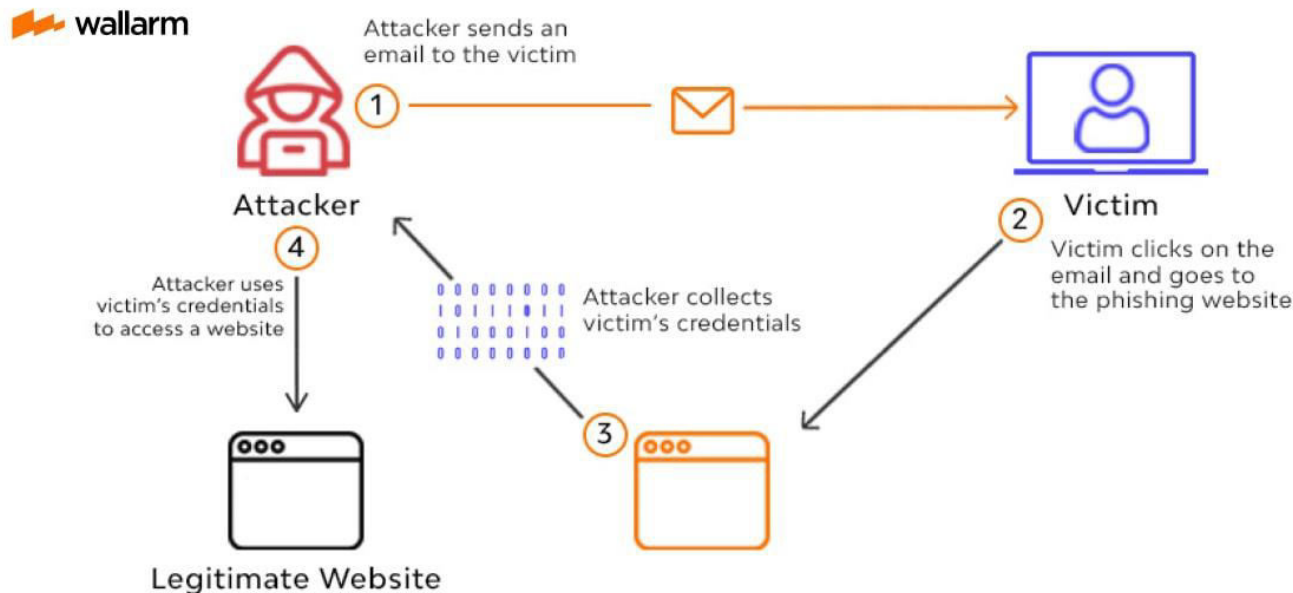
Find information about below mentioned trojan types

- Backdoor Trojans
- DDoS Trojans
- Data Sending Trojans
- Destructive Trojans
- Proxy Trojans
- Security software disabler Trojans



Phishing Attack

Phishing attacks are one of the most prominent widespread types of cyberattacks. It is a type of social engineering attack wherein an attacker impersonates to be a trusted contact and sends the victim fake mails.



Password Attack

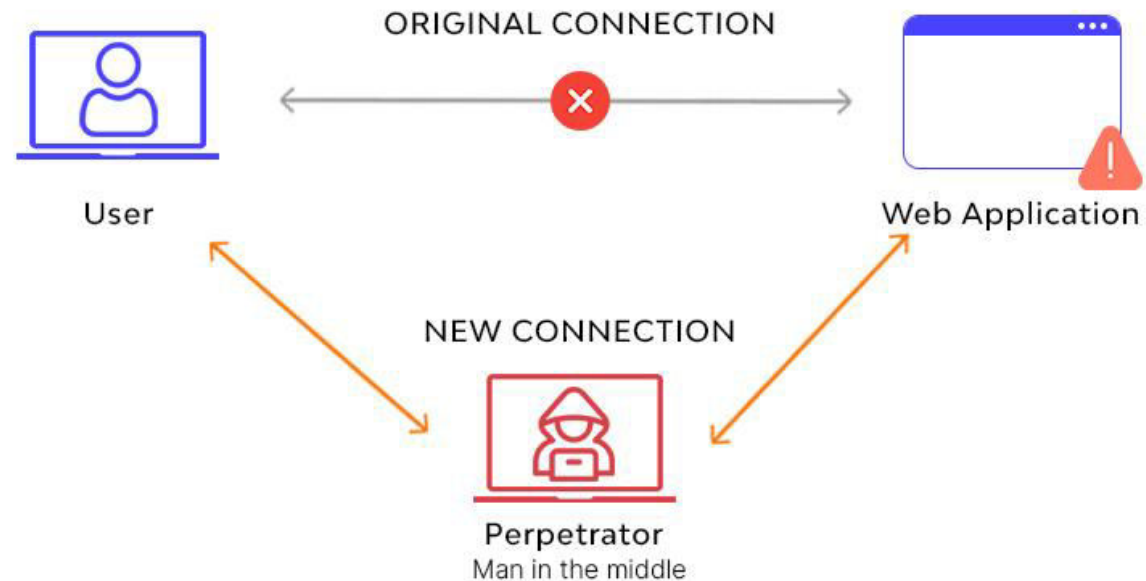
It is a form of attack wherein a hacker cracks your password with various programs.

Example Tools :- Aircrack, Cain, Abel



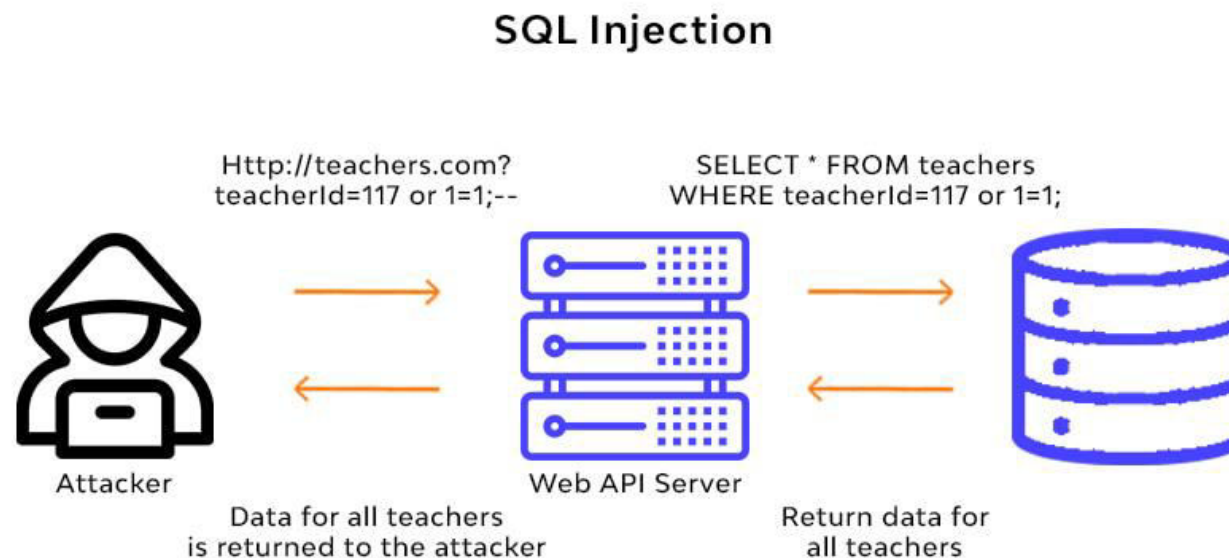
Man-in-the-Middle Attack

A Man-in-the-Middle Attack (MITM) is also known as an eavesdropping attack. In this attack, an attacker comes in between a two-party communication, i.e., the attacker hijacks the session between a client and host. By doing so, hackers steal and manipulate data.



SQL Injection Attack

A Structured Query Language (SQL) injection attack occurs on a database-driven website when the hacker manipulates a standard SQL query. It is carried by injecting a malicious code into a vulnerable website search box, thereby making the server reveal crucial information.



Denial-of-Service Attack (DOS)

A Denial-of-Service Attack is a significant threat to companies. Here, attackers target systems, servers, or networks and flood them with traffic to exhaust their resources and bandwidth.

Distributed Denial-of-Service(DDOS)

when attackers use multiple compromised systems to launch this attack.

Denial-of-Service Attack (DOS)



Other threats

- Lone Wolf Hackers
- Script Kiddies
- Cyber-Terrorists / Hacktivists



Lone Wolf Hackers

Lone wolf hackers tend to be individuals who try to access systems simply because they can.



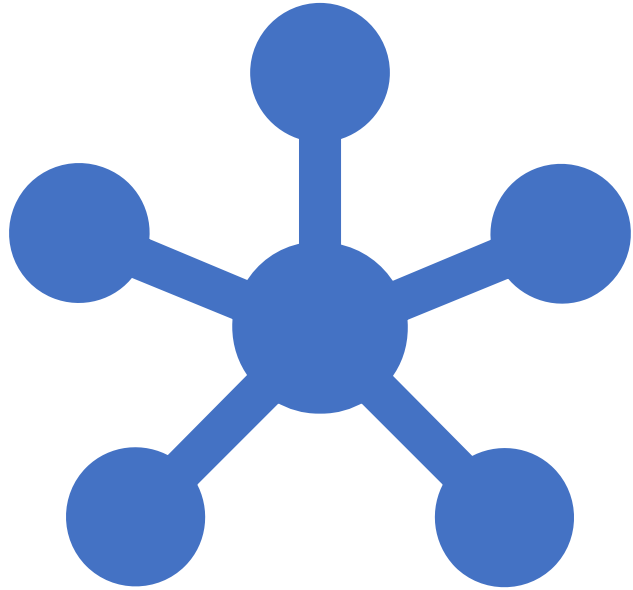
Basically, these type of hacker does not have any intention of doing any political or financial gain but these people may be motivated by revenge

Script Kiddies

Script kiddies tend to be novices. Instead of creating their own malicious code, they are usually relying heavily on scripts created by others.

These attackers are not highly experienced or skilled.





Protection Measures

Evolving Detection Measures

- In response to the malware threat, a whole industry has emerged around antivirus solutions
- Scrutinize the emails/application you receive/install.
- Update your passwords regularly.
- With the rise of crimeware, the security industry's activities move increasingly towards a policing role
- Be mindful of the security of the website you are using. Use encryption on your devices.
- Refrain from using public Wi-Fi networks.

Evolving Detection Measures

- Use an Intrusion detection system, as they design it to detect unauthorized access to a network.
- Run a traffic analysis to identify malicious traffic.
- Outsource DDoS prevention to cloud-based service providers
- Awareness Training: Educate employees about cybersecurity.
- Strong Passwords: Encourage robust, unique credentials
- Endpoint Security: Use antivirus and firewalls
- Patch Management: Regularly update software and systems
- Backup Plans: Ensure regular data backups
- Incident Response Plans: Develop and test plans for attacks

Evolving Detection Measures

- Change Default Credentials: Use strong, unique passwords and change default login credentials immediately after setup.
- Regular Firmware Updates: Keep IoT devices updated with the latest patches. If a manufacturer doesn't provide updates, consider replacing that device with a more secure model.
- Network Segmentation: Isolate IoT devices on a separate VLAN or guest network to limit lateral movement if compromised.
- Disable Unused Services: Turn off remote access and other services not actively needed.
- Monitor Network Traffic: Use intrusion detection/prevention systems (IDS/IPS) or advanced router/firewall logs to spot suspicious traffic.

Evolving Detection Measures

- Password Attack
- Do not have any password hints in the open.
- Abstain from using the same password for multiple websites or accounts.
- Detection methods have evolved in order to keep pace with malware behaviour:
 - Signature-based
 - Heuristics
 - Reputation (Cloud) based detection

Regulatory Compliance



GDPR

Protecting personal data of EU citizens.



HIPAA

Safeguarding medical information in the healthcare sector.



PCI DSS

Securing payment card data in the financial industry.

Regulatory Compliances in Sri Lanka ?

Regulatory Compliances in Sri Lanka

- **Information and Communication Technology Act (ICTA):**

The primary legislation governing IT activities in Sri Lanka, establishing the ICTA as the governing body.

- **Computer Crimes Act:**

Covers cybercrime offenses like unauthorized access to computer systems, data theft, and malicious code distribution.

- **Electronic Transactions Act:**

Regulates electronic transactions, digital signatures, and related legal aspects.

- **Data Protection Act:**

Focuses on the protection of personal data, requiring organizations to implement appropriate security measures when handling sensitive information.

Activity

What are the countermeasures that you can take in order to protect your systems





Thank You

Introduction to IoT

Department of Network and Security

Sensors and Actuators



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Outline

- **Introduction to Transducers**
- **Introduction to Sensors**
- **Reading Sensor Data Sheets and Interfacing Sensors**
- **Introduction to Actuators**
- **Reading Actuator Data Sheets and Interfacing Actuators**



Introduction to Transducers

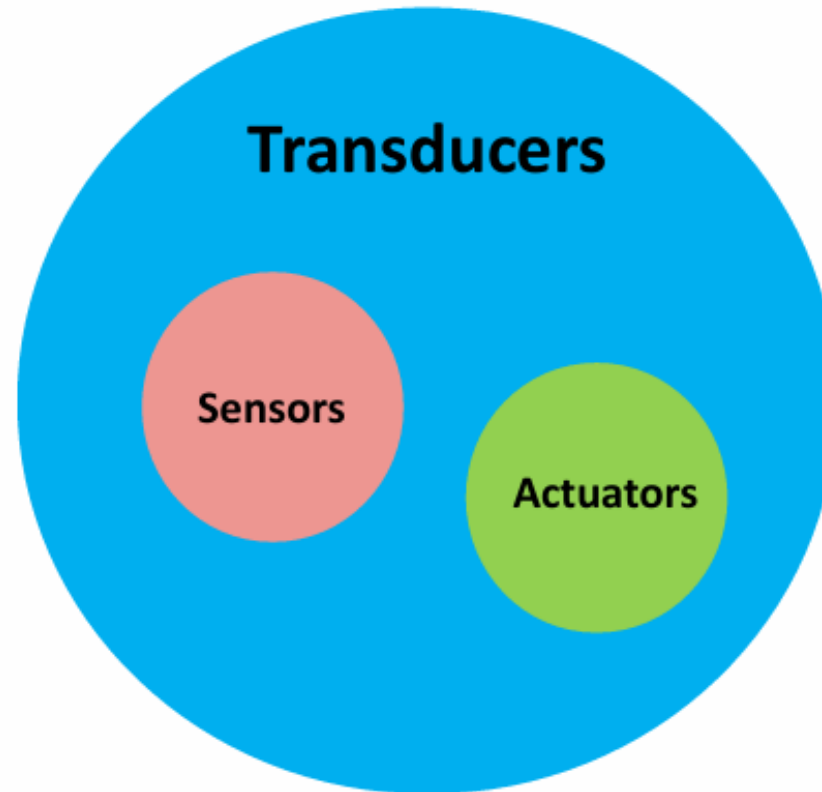
What is a Transducer ?

Transducers are devices that **convert one form of energy into another**, allowing physical quantities like temperature, pressure, sound, or light to be converted into electrical signals that can be measured, monitored, and processed. Transducers are essential in many fields, including automation, instrumentation, medical devices, and IoT applications.



What is a Transducer ?

Transducer is a term collectively used for both sensors and actuators.



Introduction to Sensors

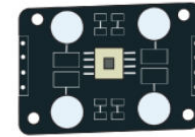
What is a Sensor ?

A sensor is an element in a mechatronic or measurement system that detects the magnitude of a physical parameter and changes it into a signal that can be processed by the system.

Monitoring and control systems require sensors to measure physical quantities such as position, distance, force, strain, temperature, vibration, and acceleration.



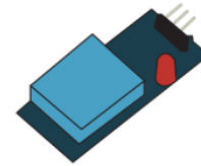
Metal Sensor



Color Sensor



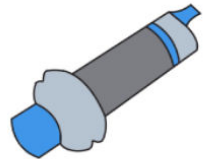
Light Sensor



Humidity Sensor



Temperature Sensor



Proximity Sensor

What is a Sensor ? *Cont`d...*

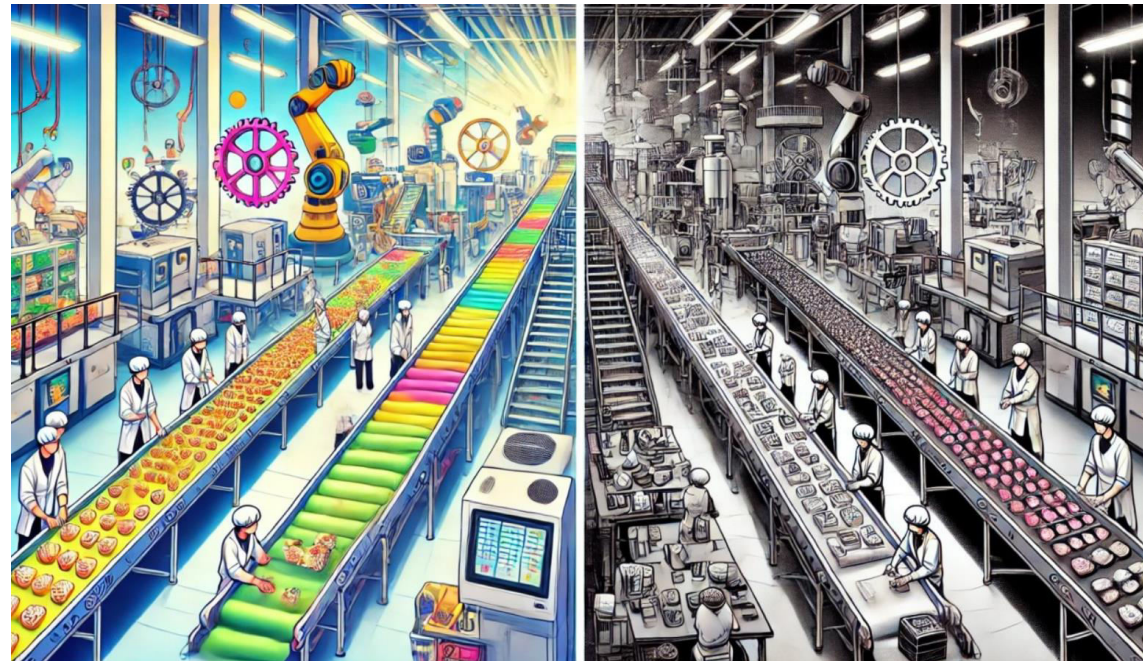
According to Instrument Society of America, “**a sensor is a device that provides usable output in response to a specified quantity which is measured.**”

The word sensor is derived from the original meaning ‘**to perceive**’

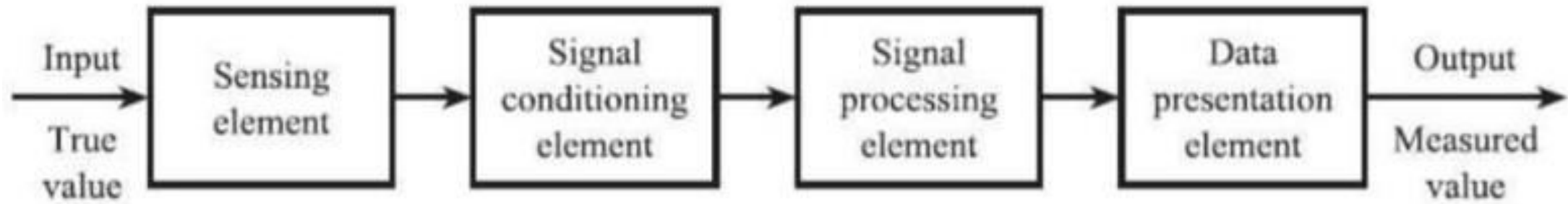
Need for Sensors

Sensors are pervasive (universal). They are embedded in automobiles, airplanes, cellular telephones, radios, chemical plants, industrial plants and countless other applications.

Without the use of sensors, there would be **no automation !!**



Components of Measuring System



Components of Measuring System *Cont`d...*

Sensing Element: It senses or detects the input and often converts it into a form that can be further processed.

E.g., Thermocouples for temperature, and LDRs (Light Dependent Resistors) for light.

Signal Conditioning Element: improves signal accuracy, adjusts levels to prevent noise interference, and transforms the signal to a compatible format.

E.g. **Amplifiers:** Increase weak signals.

Filters: Remove unwanted noise or interference.

Analog-to-Digital Converters (ADC): Convert analog signals to digital signals for digital processing.

Signal Processing Element

Data Presentation Element



Components of Measuring System *Cont`d...*

Signal Processing Element: The processing unit interprets, analyzes, and processes the signal data to make sense of it. Can be a simple microcontroller or a more advanced processor, depending on the system complexity.

Data Presentation Element: The display unit presents the processed data in a human-readable form

E.g. Digital Displays: LCD or LED screens

 Analog Meters: Moving-coil meters for visual indicators.

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Criteria to Choose Sensors

Environmental Factors	Economic Factors	Sensor Characteristics
Temperature range	Cost	Sensitivity
Humidity effects	Availability	Range
Corrosion	Lifetime	Stability
Size		Repeatability
Overrange protection		Linearity
Susceptibility to EM interferences		Error
Ruggedness		Response time
Power consumption		Frequency response
Self-test capability		

Criteria to Choose Sensors *Cont`d...*

Type of Sensing: The parameter that is being sensed like temperature or pressure.

Operating Principle: The principle of operation of the sensor.

Power Consumption: The power consumed by the sensor will play an important role in defining the total power of the system.

Accuracy: The accuracy of the sensor is a key factor in selecting a sensor.

Criteria to Choose Sensors *Cont`d...*

Environmental Conditions: The conditions in which the sensor is being used will be a factor in choosing the quality of a sensor.

Cost: Depending on the cost of application, a low-cost sensor or high-cost sensor can be used.

Resolution and Range: The smallest value that can be sensed and the limit of measurement are important.

Calibration and Repeatability: Change of values with time and ability to repeat measurements under similar conditions

Basic Requirements of a Sensors *Cont`d...*

- **Range:** It indicates the limits of the input in which it can vary. In case of temperature measurement, a thermocouple can have a range of 25 –250 0C.
- **Accuracy:** It is the degree of exactness between actual measurement and true value. Accuracy is expressed as percentage of full range output.
- **Sensitivity:** Sensitivity is a relationship between input physical signal and output electrical signal. It is the ratio of change in output of the sensor to unit change in input value that causes change in output.

Basic Requirements of a Sensors *Cont`d...*

- **Stability:** It is the ability of the sensor to produce the same output for constant input over a period of time.
- **Repeatability:** It is the ability of the sensor to produce same output for different applications with same input value.
- **Response Time:** It is the speed of change in output on a stepwise change in input.
- **Linearity:** It is specified in terms of percentage of nonlinearity. Nonlinearity is an indication of deviation of curve of actual measurement from the curve of ideal measurement.

Basic Requirements of a Sensors *Cont`d...*

- **Ruggedness:** It is a measure of the durability when the sensor is used under extreme operating conditions.
- **Hysteresis** in a sensor refers to the difference in output readings when a measured parameter (such as temperature, pressure, or displacement) is approached from increasing versus decreasing directions. This effect is typically due to the sensor's internal properties, where the output lags behind the input, resulting in different readings for the same input level depending on the previous state.

E.g. Consider a **pressure sensor**. If you increase the pressure from 0 to 50 psi and then decrease it back to 0, the output might follow different paths for increasing and decreasing pressure, giving different readings for the same pressure level.

Classification of Sensors

Classification based on **stimuli** that is being sensed

- **Acoustic:** Wave, spectrum and wave velocity.
- **Electric:** Current, charge, potential, electric field, permittivity and conductivity.
- **Magnetic:** Magnetic field, magnetic flux and permeability

Classification of Sensors

Classification based on **stimuli** that is being sensed

- **Thermal:** Temperature, specific heat and thermal conductivity.
- **Mechanical:** Position, acceleration, force, pressure, stress, strain, mass, density, momentum, torque, shape, orientation, roughness, stiffness, compliance, crystallinity and structural.
- **Optical:** Wave, wave velocity, refractive index, reflectivity, absorption and emissivity

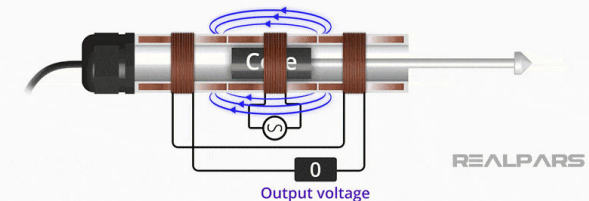
Classification of Sensors *Cont`d...*

Classification based on **Application**

1. Displacement Sensors

- **Purpose:** Measure the distance moved by an object from a reference point.
- **Applications:** Typically used for linear or angular displacement measurements in manufacturing, assembly, and quality control.
- **Types:**
 - **LVDT (Linear Variable Differential Transformer):**
 - Measures linear displacement with high accuracy.
 - Operates by detecting changes in inductance as the core moves within coils.
 - **Application:** CNC machines, hydraulic systems.

Electromechanical passive inductive transducer



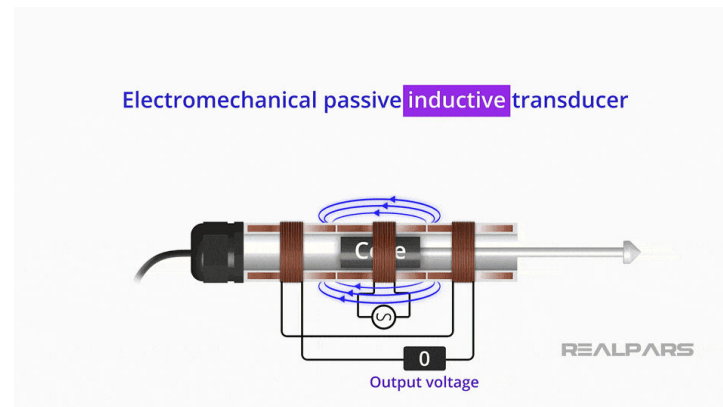
Classification of Sensors *Cont`d...*

Classification based on **Application**

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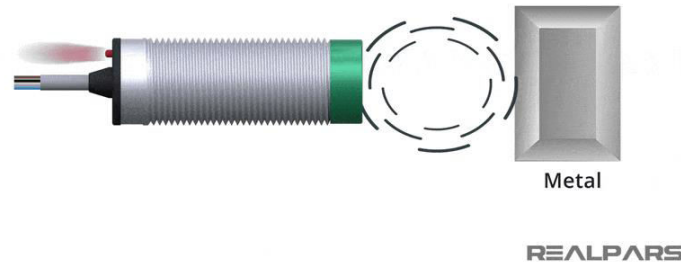
Classification of Sensors *Cont`d...*

Classification based on **Application**

1. Displacement Sensors

Capacitive Displacement Sensors:

- Measure changes in capacitance caused by displacement.
- Non-contact, suitable for small distances and high precision.
- **Application:** Semiconductor manufacturing, micro-positioning.



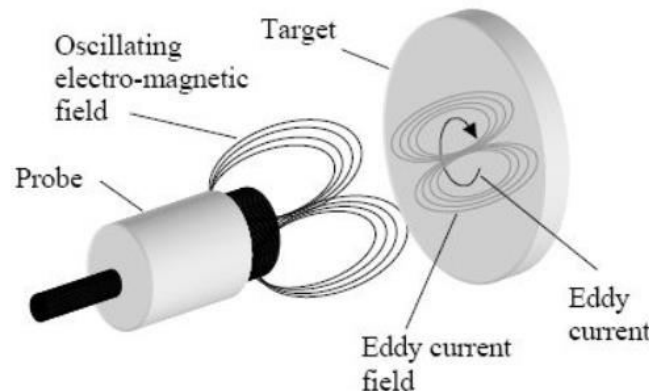
Classification of Sensors *Cont`d...*

Classification based on **Application**

1. Displacement Sensors

Inductive Displacement Sensors:

- Use a coil and magnetic core to measure displacement based on changes in inductance.
- Robust and used for larger displacements.
- **Application:** Automotive systems, assembly lines.



Classification of Sensors *Cont`d...*

Classification based on **Application**

2. Position Sensors

Purpose: Detect the specific position or location of an object.

Applications: Widely used in applications requiring precise positioning or alignment, such as in robotics, automotive steering, and camera stabilizers.

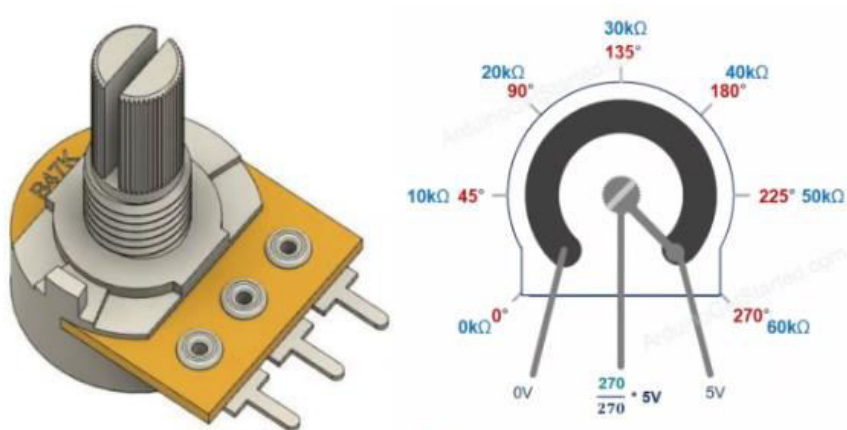
Classification of Sensors *Cont`d...*

Classification based on **Application**

2. Position Sensors

Potentiometers

- Use a resistive track and a movable contact to measure angular or linear position.
- Simple and cost-effective, but subject to wear.
- Application: Joysticks, volume controls, robotics.



Classification of Sensors *Cont`d...*

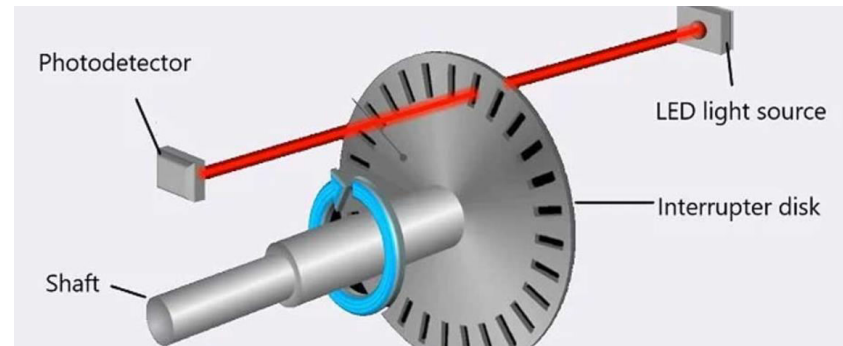
Classification based on **Application**

2. Position Sensors

Encoders :

1. Rotary Encoders: Measure rotational position; can be incremental (track changes) or absolute (track exact position).
2. Linear Encoders: Track linear movement, often with optical or magnetic strips.

Application: CNC machines, robotic arms, motor feedback.



Classification of Sensors *Cont`d...*

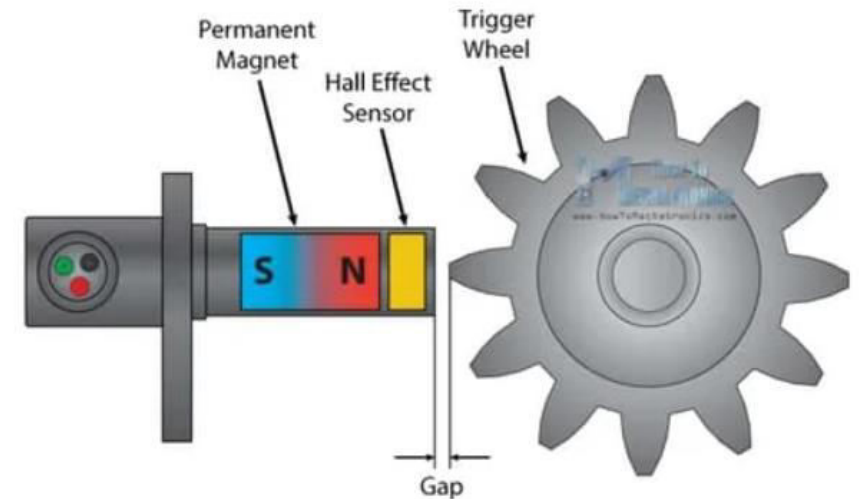
Classification based on **Application**

2. Position Sensors

Hall Effect Sensors:

- Use magnetic fields to detect position changes, particularly useful in detecting rotational position.
- Non-contact, robust in dusty or dirty environments.
- **Application:** Automotive throttle and pedal position sensing, wheel speed measurement.

HALL EFFECT SENSOR



Classification of Sensors *Cont`d...*

Classification based on **Application**

3. Proximity Sensors

Purpose: Detect the presence or absence of an object, often without making physical contact.

Applications: Commonly used in security systems, automation, and object detection in manufacturing lines.

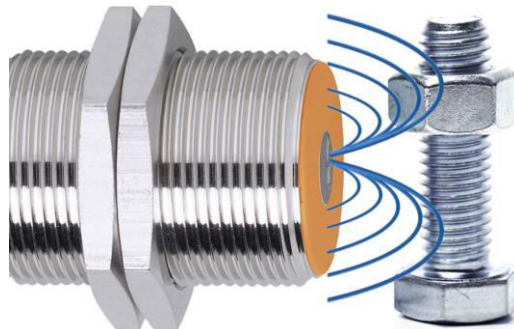
Classification of Sensors *Cont`d...*

Classification based on **Application**

3. Proximity Sensors

Inductive Proximity Sensors:

- Detect metallic objects by creating an electromagnetic field that changes when a metallic object enters it.
- Non-contact, durable, and suitable for harsh environments.
- **Application:** Metal detection, automotive assembly.



Classification of Sensors *Cont`d...*

Classification based on **Application**

3. Proximity Sensors

Capacitive Proximity Sensors:

- Sense changes in capacitance caused by objects, detecting both metallic and non-metallic materials.
- Sensitive to moisture and dust, so often used in clean environments.
- **Application:** Liquid level detection, packaging.



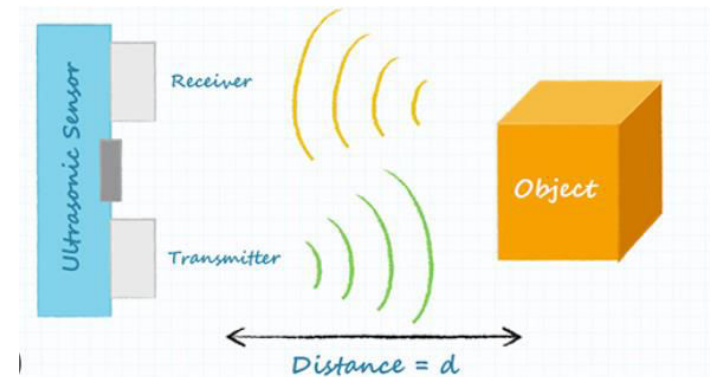
Classification of Sensors *Cont`d...*

Classification based on **Application**

3. Proximity Sensors

Ultrasonic Proximity Sensors:

- Emit ultrasonic waves and measure the echo to detect objects and measure distance.
- Effective for long-range detection and non-contact measurement.
- **Application:** Object detection, parking assistance, industrial distance sensing.



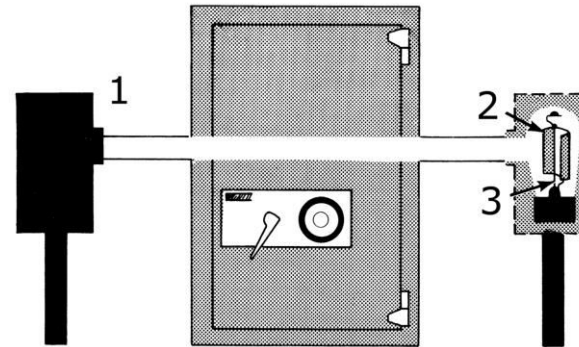
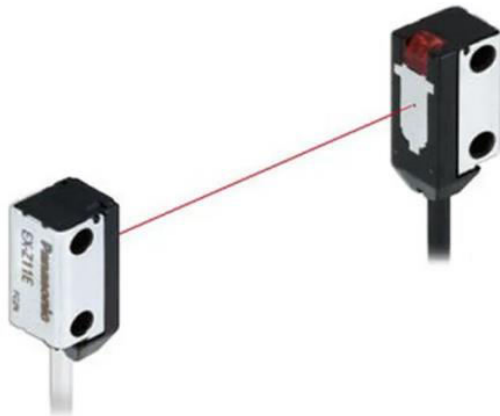
Classification of Sensors *Cont`d...*

Classification based on **Application**

3. Proximity Sensors

Photoelectric Sensors:

- Use light (infrared or visible) to detect the presence of an object. Includes **reflective**, **through-beam**, and **diffuse** types.
- Versatile and used for detecting a variety of objects.
- **Application:** Conveyor systems, presence detection, door sensors.



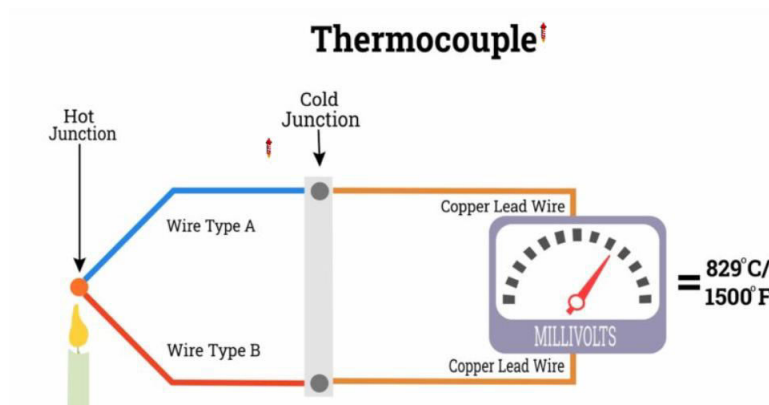
Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Thermocouples

- Working Principle: Based on the **Seebeck effect**, where a voltage is generated between two dissimilar metals when they experience a temperature difference.
- Temperature Range: -200°C to 1750°C , depending on the thermocouple type.
- Advantages: Wide range, rugged, fast response.
- Applications: Industrial ovens, kilns, engines, and exhaust systems.



Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Resistance Temperature Detectors (RTDs)

- Working Principle: RTDs are based on the principle that the electrical resistance of metals (like platinum) increases with temperature.
- Temperature Range: -200°C to 850°C .
- Advantages: High accuracy, stable, linear output.
- Applications: Laboratory measurements, HVAC systems, and industrial process control.



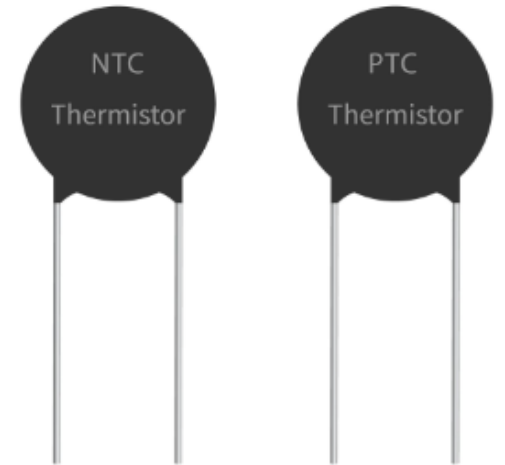
Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Thermistors

- Working Principle: Thermistors are temperature-sensitive resistors, typically made from ceramic or polymer. NTC (Negative Temperature Coefficient) thermistors decrease in resistance with increasing temperature, and PTC (Positive Temperature Coefficient) thermistors increase in resistance.
- Temperature Range: -50°C to 150°C (typically).
- Advantages: High sensitivity, fast response, cost-effective.
- Applications: Consumer electronics (e.g., temperature control in appliances), automotive sensors, and medical devices.



Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Semiconductor Temperature Sensors

- Working Principle: Utilize the properties of semiconductors, like diodes and transistors, where voltage or current characteristics change with temperature.
- Temperature Range: -55°C to 150°C .
- Advantages: Highly linear, low cost, integrated circuits available.
- Applications: Microcontrollers, computers, battery management systems, and other electronics.



Temperature Sensor

Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Infrared (IR) Temperature Sensors

- Working Principle: Measure the infrared radiation emitted by an object to determine its temperature, making them ideal for non-contact temperature measurements.
- Temperature Range: -70°C to 1000°C .
- Advantages: Non-contact, fast response, suitable for moving or hazardous objects.
- Applications: Food processing, medical thermometers, HVAC systems, and industrial equipment monitoring.



Classification of Sensors *Cont`d...*

Classification based on **Application**

4. Temperature Sensors

Thermal Imaging Sensors (Infrared Cameras)

- Working Principle: Use a matrix of IR sensors to capture a detailed temperature profile over an area, creating a thermal image.
- Temperature Range: Varies based on the sensor but typically capable of measuring a broad range.
- Advantages: Non-contact, provides detailed temperature distribution.
- Applications: Building diagnostics, medical imaging, firefighting, predictive maintenance in machinery.



Classification of Sensors *Cont`d...*

Classification based on **Application**

5. Light Sensors

- Photo Diode
- Phototransistor
- Light Dependent Resistor

6. Velocity and Motion

- Pyroelectric Sensors
- Tachogenerator
- Incremental encoder

Classification of Sensors *Cont`d...*

Classification based on **Application**

7. Fluid Pressure

- Diaphragm Pressure Gauge
- Tactile Sensor
- Piezoelectric Sensors
- Capsules, Bellows, Pressure Tubes

6. Liquid Flow and Level

- Turbine Meter
- Orifice Plate and VenturiTube

Classification of Sensors *Cont`d...*

Classification based on **Application**

8. IR Sensor

- Infrared Transmitter and Receiver Pair

9. Force

- Strain Gauge
- Load Cell

10. Touch Sensors

- Resistive Touch Sensor
- Capacitive Touch Sensors

Classification of Sensors *Cont`d...*

Classification based on **Application**

11. UV Sensors

- Ultraviolet Light Detector
- Photo Stability Sensors
- UV Photo Tubes
- Germicidal UV Detectors

12. Chemical Sensors

Reading Sensor Datasheets and Interfacing Sensors

Data Sheet Reading

Reading a sensor datasheet is essential to understand the sensor's specifications, capabilities, and limitations for proper integration into a project or system.

Features

- Lists key attributes and functions, such as measurement range, accuracy, power consumption, and unique benefits (like high sensitivity, digital output).

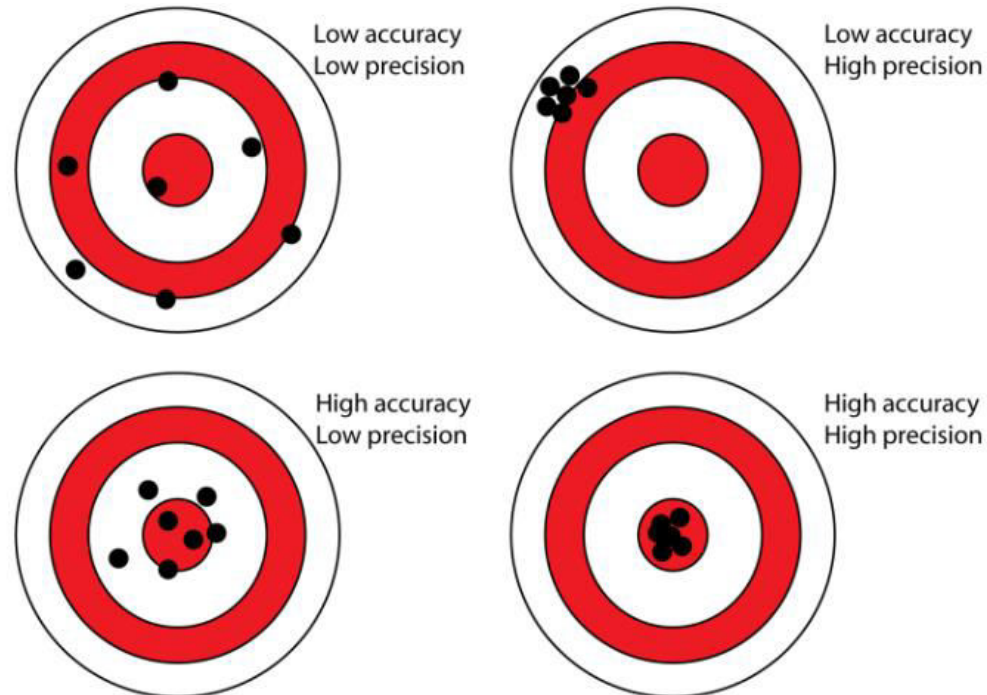
Electrical Characteristics

- Supply Voltage: Indicates the operating voltage range (e.g., 3.3V to 5V) and required current.
- Current Consumption: The amount of current the sensor draws during operation, which is essential for power-sensitive applications.
- Output Type: Specifies whether the sensor has an analog, digital (I2C, SPI), or PWM output.

Data Sheet Reading *Cont`d...*

Accuracy and Precision

- Accuracy: Describes how close the sensor's reading is to the actual value. It is often presented as a tolerance or $\pm$ percentage of the measured range.
- Precision (Repeatability): Indicates the sensor's ability to provide consistent readings under the same conditions.



Data Sheet Reading *Cont`d...*

Measurement Range

- Defines the minimum and maximum values the sensor can measure reliably.
- Example: A temperature sensor might have a range from -40°C to 125°C.

Resolution

- Refers to the smallest detectable change in the measured parameter that the sensor can resolve.
- Often given in bits (e.g., 12-bit resolution for an ADC).

Response Time

Data Sheet Reading *Cont`d...*

Operating Conditions

- Temperature Range: Specifies the environmental temperature range in which the sensor can operate safely.
- Humidity Range: Some sensors specify acceptable humidity levels, especially if they are humidity-sensitive.
- Storage Conditions: Safe temperature and humidity ranges for storing the sensor.

Mechanical Dimensions and Mounting

- Provides physical dimensions and mounting instructions for integrating the sensor onto a PCB or other hardware.

Pin Configuration and Connections

- A diagram of the sensor's pin layout, including each pin's function (e.g., VCC, GND, SDA, SCL for I2C sensors).

Data Sheet Reading *Cont`d...*

Calibration and Compensation Information

- Some sensors require calibration, which may be user-performed or factory-provided. The datasheet will mention if calibration is needed or included.
- **Sensor calibration** is the process of adjusting a sensor's output to ensure its accuracy and reliability in measuring a specific parameter (e.g., temperature, pressure, distance). Calibration aligns the sensor's readings with a known reference standard, minimizing errors and improving precision.

Data Sheet Reading *Cont`d...*

Activity

Reading the HC-SR04 Ultrasonic Sensor Datasheet and Listing Key Specifications.

- **Working Voltage:**
- **Operating Frequency:**
- **Current Consumption:**
- **Measurement Range:**
- **Accuracy:**
- **Detection Angle:**
- **Pin Configuration**

Activity

Interface HC-SR04 Ultrasonic Sensor to a microcontroller by understanding its working principle .

Ultrasound refers to sound waves with frequencies higher than the upper audible limit of human hearing, typically above 20 kHz. Ultrasound devices generate high-frequency sound waves using a piezoelectric transducer. The waves propagate through a medium and interact with objects, producing echoes. These echoes are analyzed to extract information about the object's shape, size, and location.

Sensor Interfacing *Cont`d...*

Activity

HC-SR04 Ultrasonic Sensor

Key Features:

1. **Operating Voltage:** 5V DC
2. **Frequency:** 40 kHz (ultrasonic wave)
3. **Distance Range:**
 1. Minimum: 2 cm
 2. Maximum: 400 cm (4 meters)
4. **Accuracy:** ± 3 mm
5. **Interface:** Digital I/O pins (Trigger and Echo)
6. **Compact Size:** Approximately 45mm x 20mm x 15mm

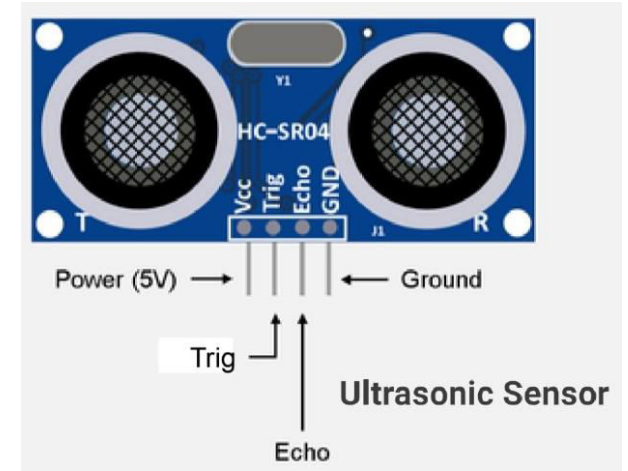
Sensor Interfacing *Cont`d...*

Activity

HC-SR04 Ultrasonic Sensor

Pin Description:

1. **VCC:** Connects to the 5V power supply.
2. **Trig:** Trigger pin, used to send a pulse to initiate the measurement. (5V 10 microseconds)
3. **Echo:** Echo pin, outputs the duration of the received echo pulse. (outputs a pulse of 150microseconds to 25 milliseconds). Echo pin will timeout after 38mS if no object is detected.
4. **GND:** Connects to the ground.

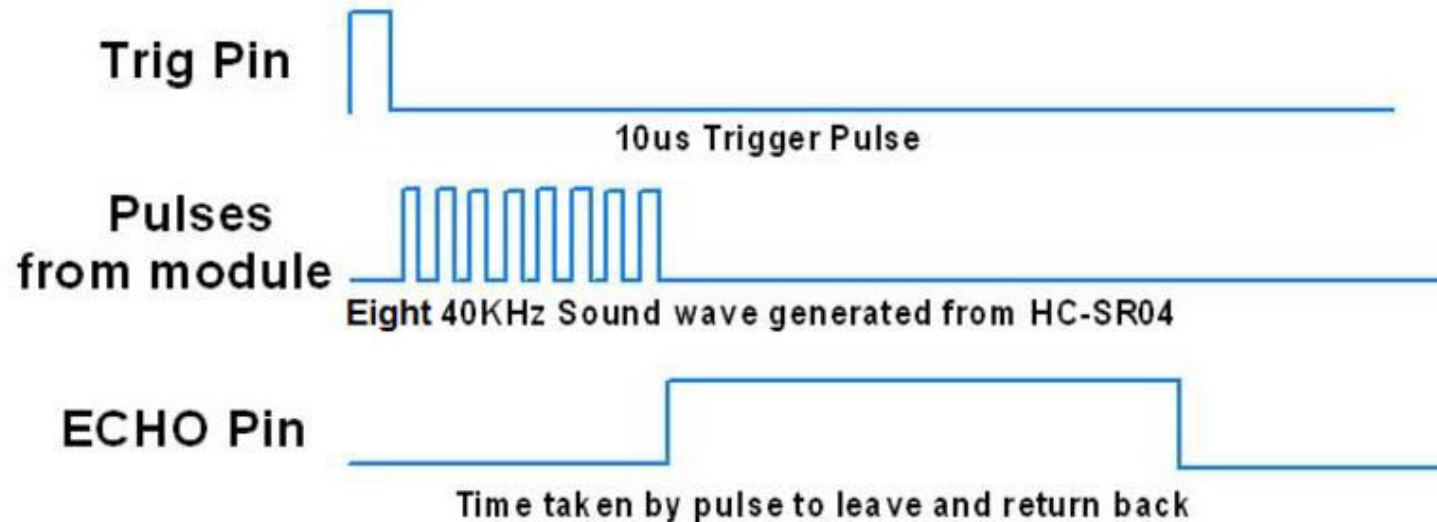


Sensor Interfacing *Cont`d...*

Activity

HC-SR04 Ultrasonic Sensor

Ultrasonic HC-SR04 module Timing Diagram



Sensor Interfacing *Cont`d...*

Activity

HC-SR04 Ultrasonic Sensor Calculating Distance

- Ultrasonic signal travels at the speed of sound at 20 Celsius (343 m/s)
 - t = time delay
 - C = speed of sound (343 m/s in dry air at 20 Celsius)
 - D = measuring distance
- Sound takes 29.154 microseconds to travel one centimeter

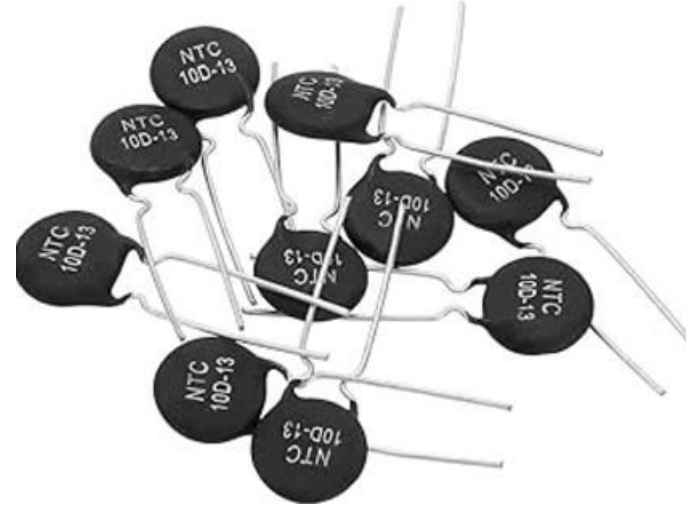
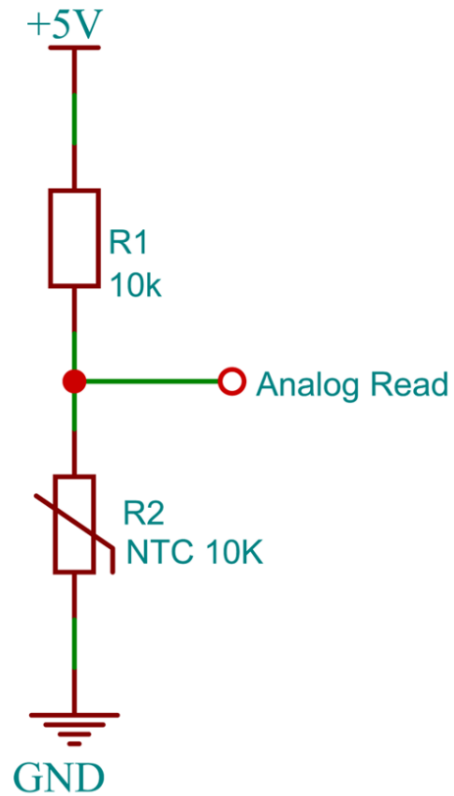
$$D = (t/2) \times C$$

$$D = (t/2) / 29.154$$

Sensor Interfacing *Cont`d...*

Activity

NTC Analog Sensor



Introduction to Actuators

What is an Actuator ?

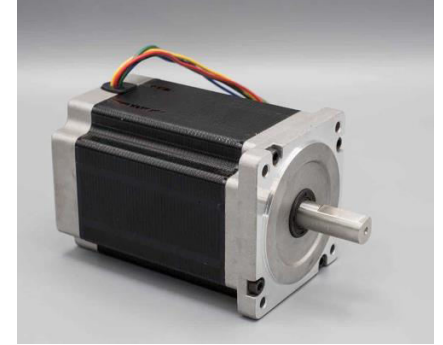
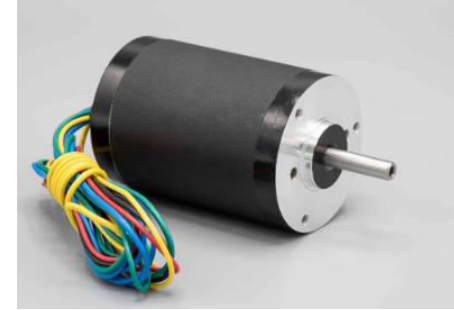
Actuators: Hardware devices that convert a controller command signal into a change in a physical parameter.

- The change is usually mechanical (e.g., position or velocity).
- An actuator is also a transducer because it changes one type of physical quantity into some alternative form(e.g. electric current to rotational speed of electric motor).

Types of Actuators

1. Electrical actuators

- Electric motors (linear or rotational)
- DC servomotors
- AC motors
- Stepper motors
- Solenoids
- Relay



2. Pneumatic actuators

Use compressed air as the driving force

3. Hydraulic actuators

Use hydraulic fluid as the driving force

Electric Motors

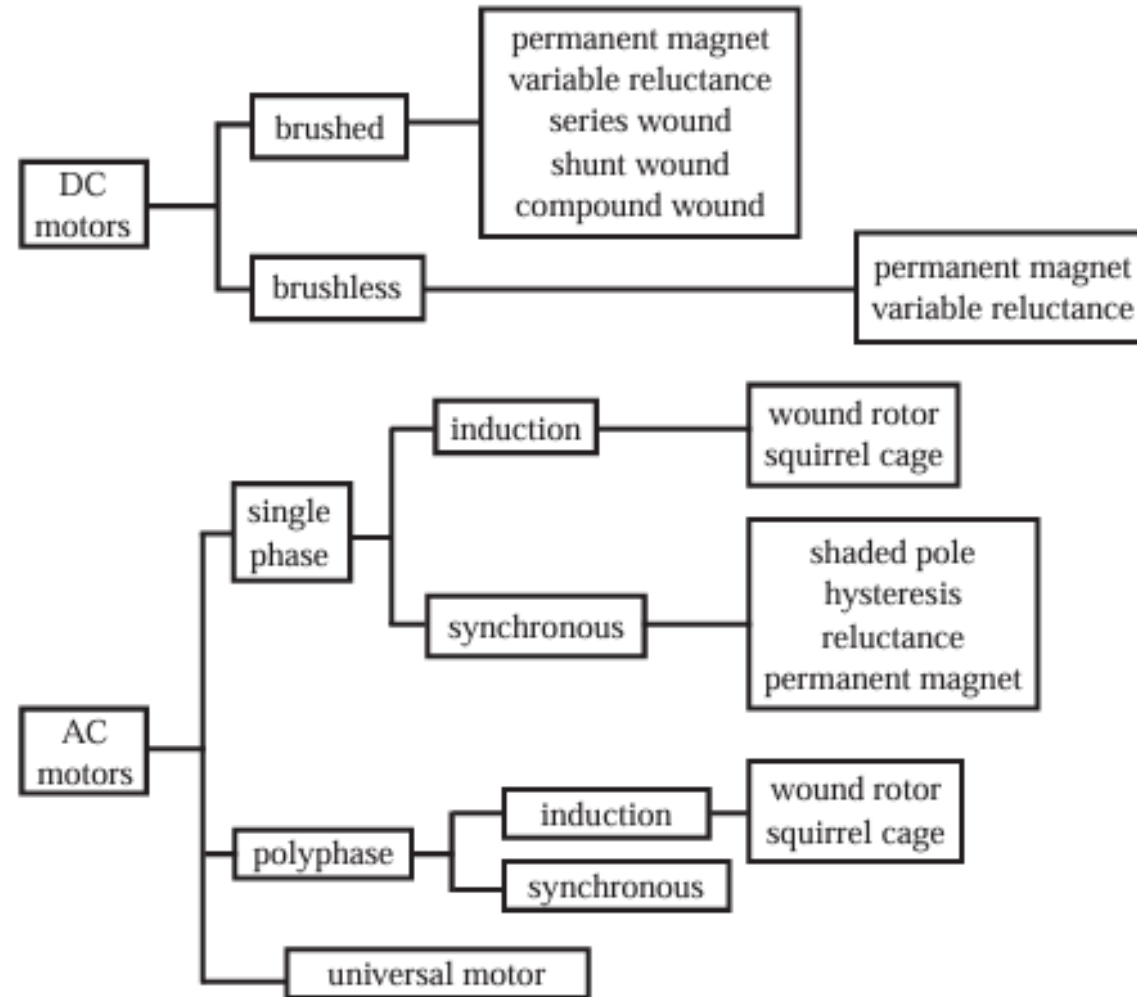


Figure 10.5 Configuration classification of electric motors.

DC Servo Motors



- A specific kind of electric motor called a servo motor is made to precisely control acceleration, velocity, and angular or linear position.
- Servo motors are unique among motor types because they have feedback control mechanisms that allow them to precisely control position within a closed-loop system.

DC Servo Motors

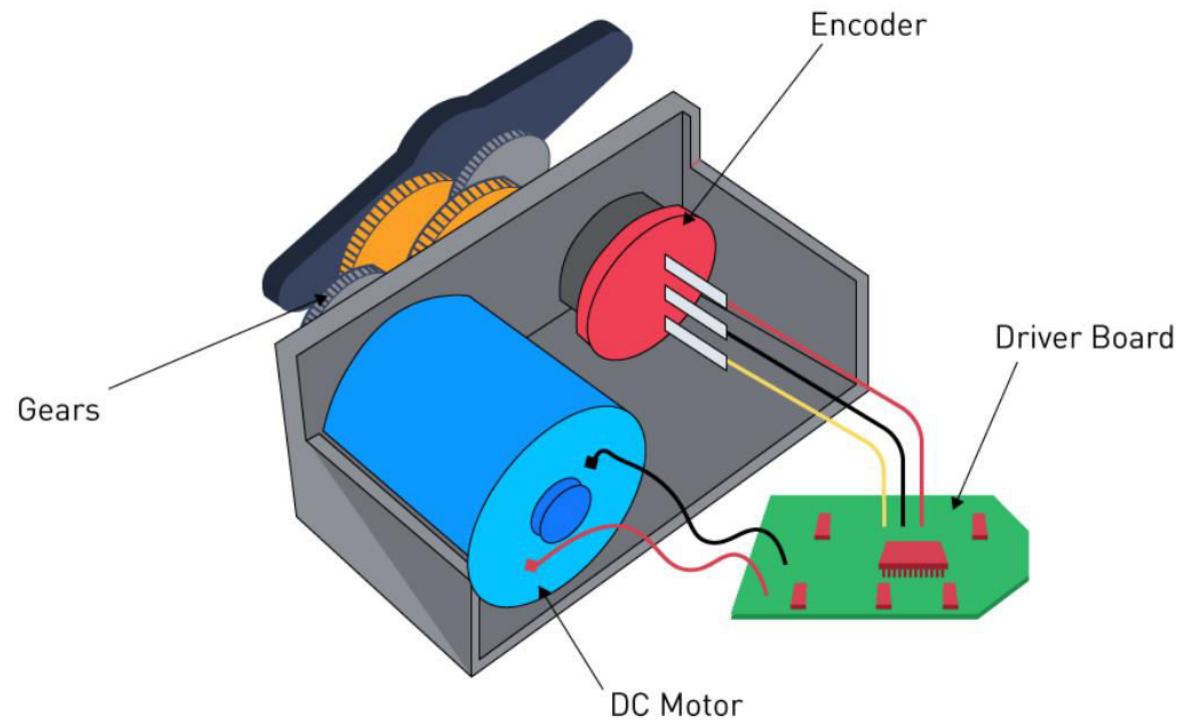
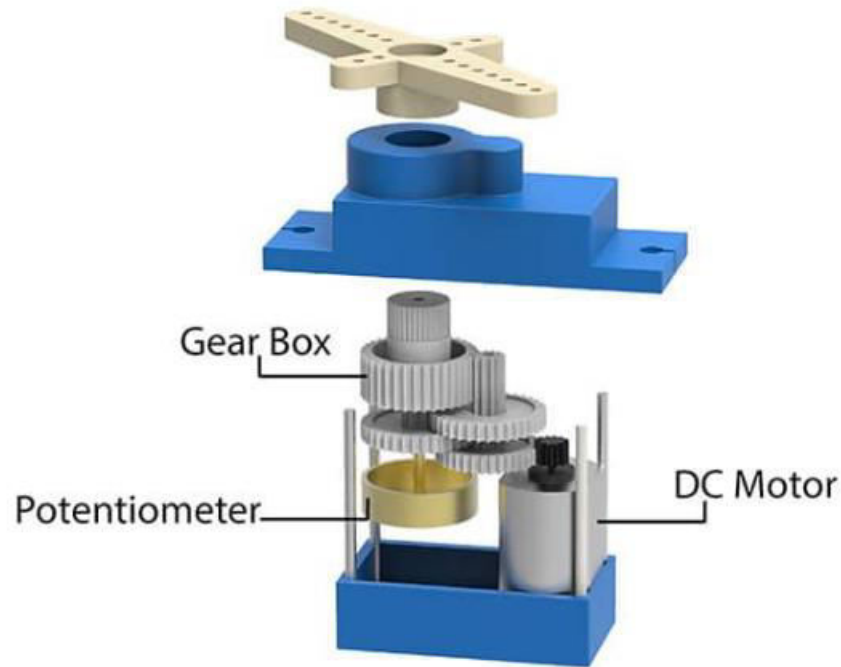


Figure 1: Servo Motor Components

- The rotation angle of the servo motor is controlled by applying a **PWM** signal to it.
- By varying the width of the PWM signal, we can change the rotation angle and direction of the motor.

DC Servo Motors *Cont`d...*



Advantages and Disadvantages

- High precision, good speed control, low-cost options, and availability in small sizes.
- Limited torque at high speeds, sensitivity to load changes, and they can be prone to overheating if overused.

DC Servo Motors *Cont`d...*

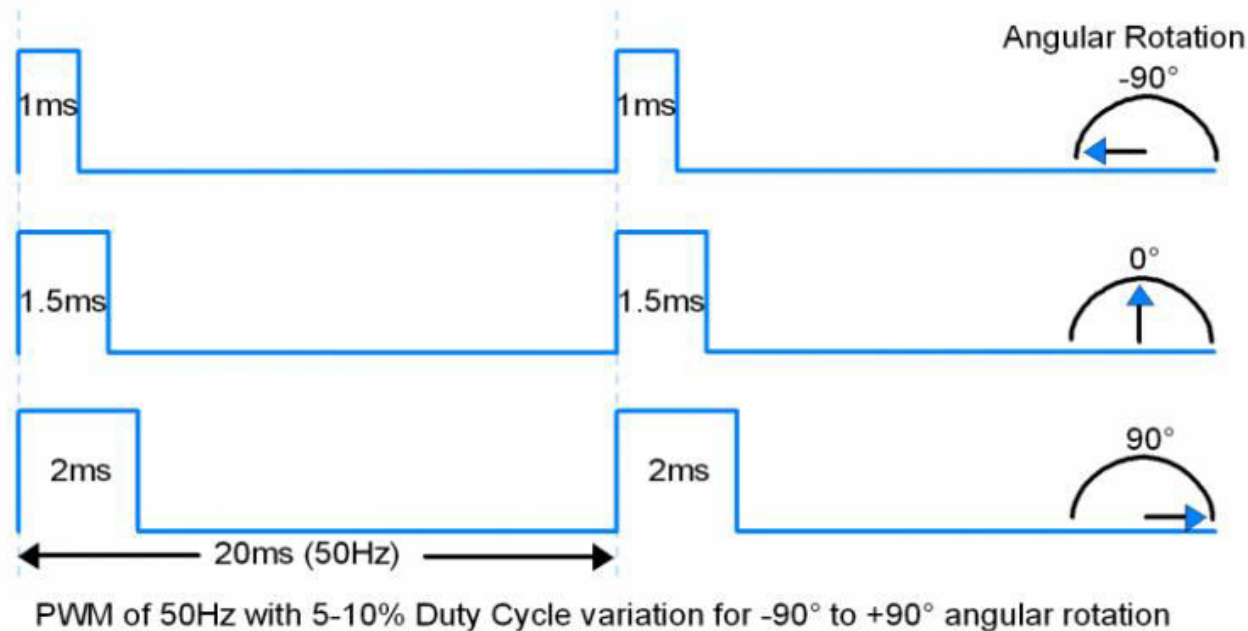


Applications

- Robotics: For precise control of arms and joints.
- CNC Machines: For accurate positioning in milling and cutting.
- Automotive: Often in braking and steering systems.

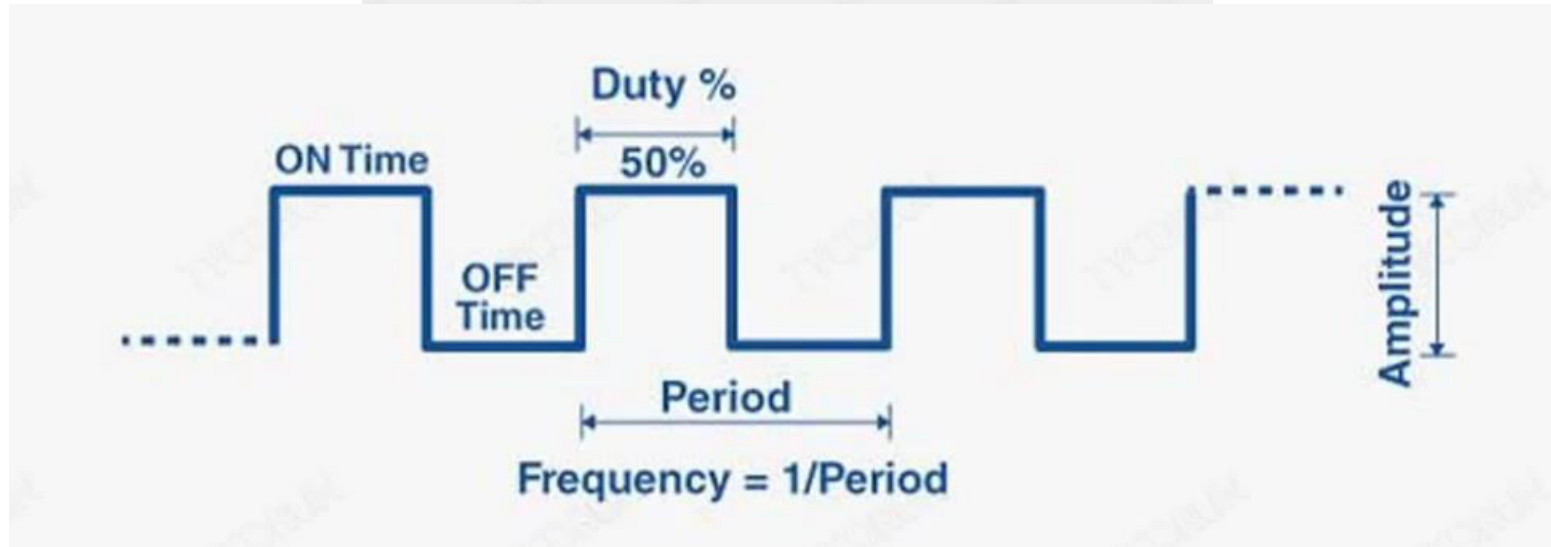
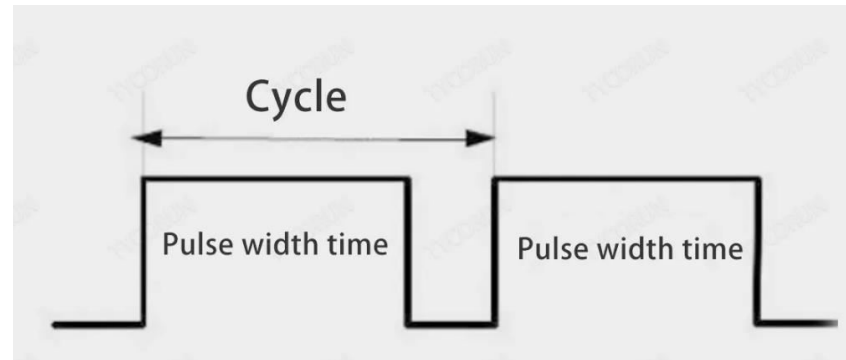
DC Servo Motors *Cont`d...*

Servo motor position is controlled by pulse width modulation (PWM) signal. The width of the control signal pulse is used to vary shaft position of servo motor. We can vary SG90 Micro servo motor angular rotation in between 0° to 180° angle with PWM signal as shown in below figure.

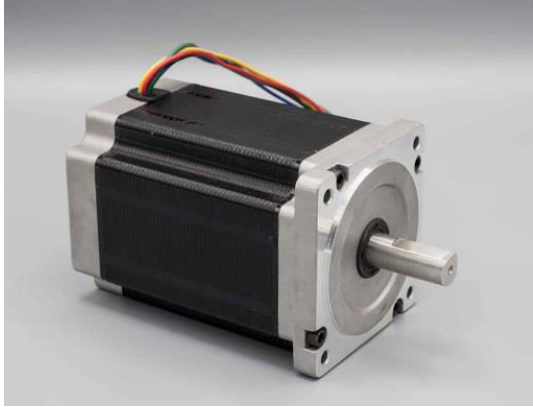


DC Servo Motors *Cont`d...*

$$\text{Duty Cycle} = \frac{T_{\text{on}}}{T_{\text{on}} + T_{\text{off}}}$$



Stepper Motors



Stepper motors are precise, electromechanical devices that convert electrical pulses into distinct mechanical steps. This makes them ideal for applications requiring accurate positioning and control.

Stepper Motors *Cont`d...*

- Stepper motor is a brushless DC motor that divides the full rotation angle of 360° into a number of equal steps.
- The motor is rotated by applying a certain sequence of control signals. The speed of rotation can be changed by changing the rate at which the control signals are applied.
- Various stepper motors with different step angles and torque ratings are available in the market.
- A microcontroller can be used to apply different control signals to the motor to make it rotate according to the need of the application.

Stepper Motors *Cont`d...*

Common Control Modes

Full-Step Mode: Moves the motor one full step at a time, usually 1.8° per step for a typical motor.

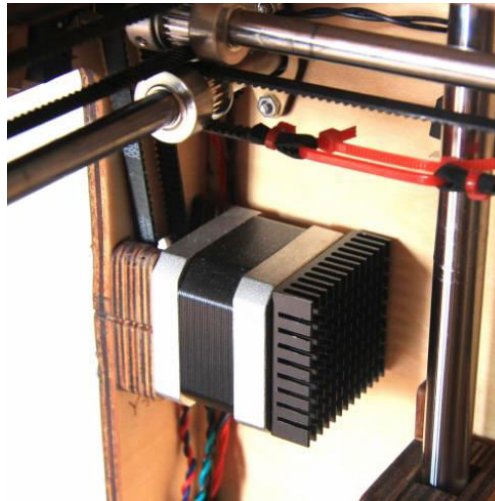
Half-Step Mode: Alternates between full and half steps, reducing each step angle (e.g., 0.9° per step) for smoother motion.

Microstepping Mode: Divides steps even further (e.g., $1/16$, $1/32$ of a step), creating extremely smooth and precise movements. Requires more complex drivers but significantly reduces noise and vibration.

Stepper Motors *Cont`d...*

Applications of Stepper Motors

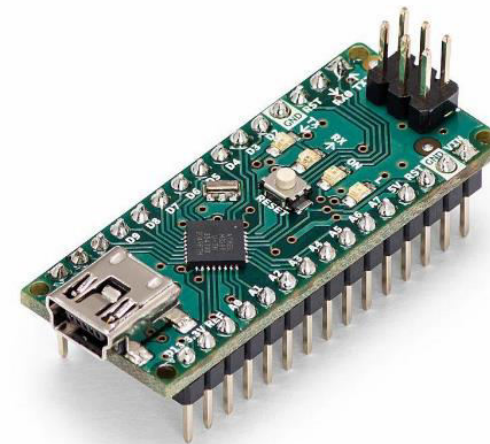
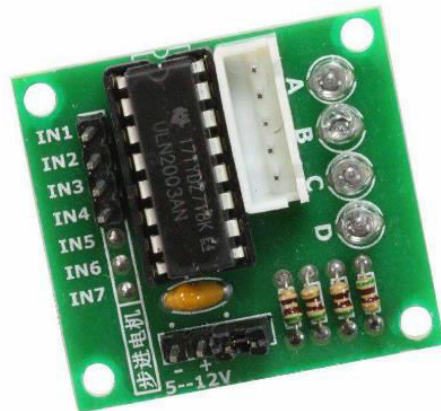
- 3D Printers and CNC Machines: For precise control over position and movement.
- Camera Gimbals: Ensures smooth, stable positioning of cameras.
- Medical Devices: For precise operations in equipment like MRI machines.



Stepper Motors *Cont`d...*

Control Stepper Motors

- **Driver Circuit:** Stepper motors require driver circuits to sequence the current through the motor windings. Stepper drivers allow for control over the mode (full, half, microstepping).
- **Pulse Generator:** Often a microcontroller or dedicated control board generates the pulses needed to move the motor.



Solenoids



Solenoids are electromagnetic devices that convert electrical energy into linear mechanical motion. They consist of a coil of wire (solenoid coil) wrapped around a movable metal core.

Applications

- Automotive: Used in car door locks, fuel injectors, and starter motors.
- Industrial Automation: Solenoids in control valves, actuators, and relays are widely used to control fluid and gas flow.

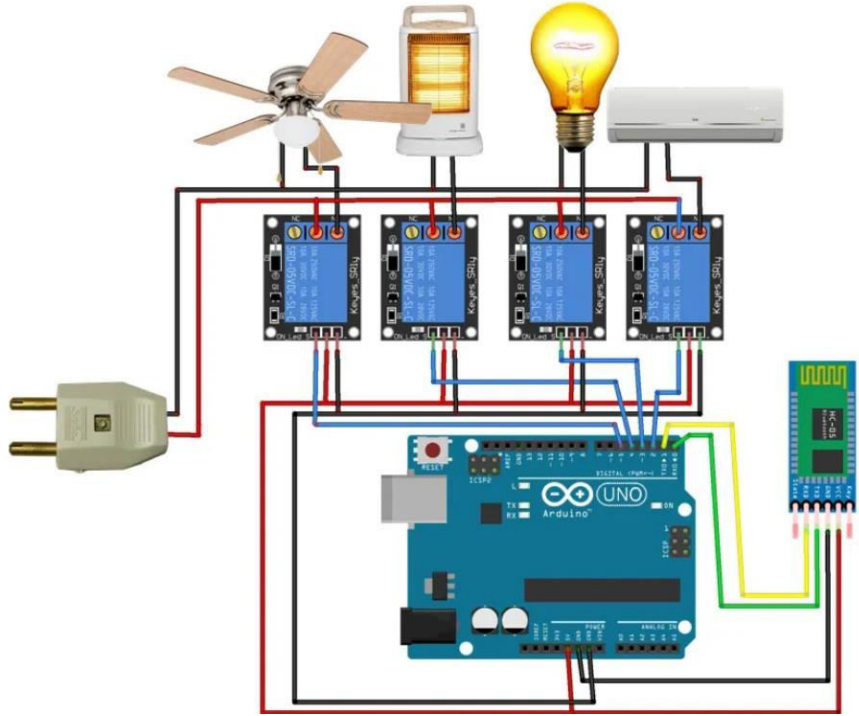
Relays



Relays are electromagnetic switches that use a small electrical current to control a larger current in a separate circuit. They are used to control high-power devices, electrically isolate circuits, or trigger multiple circuits with a single signal.

Relays Cont'd...

Applications



- Automotive: Control headlights, windshield wipers, and fuel pumps
- Industrial Automation: Used in motor control, heating elements, and conveyor systems.
- Home Appliances: Common in washing machines, refrigerators, and HVAC systems for controlling high-power components.

Reading Actuator Datasheets and Interfacing Actuators

Data Sheet Reading

Basic Specifications

Type: Describes the actuator type (linear, rotary, pneumatic, electric, hydraulic)

Operating Voltage/Current: Indicates the voltage and current needed to power the actuator.

Stroke Length (for Linear Actuators): The maximum distance the actuator can move in one direction.

Data Sheet Reading *Cont`d...*

Performance Parameters

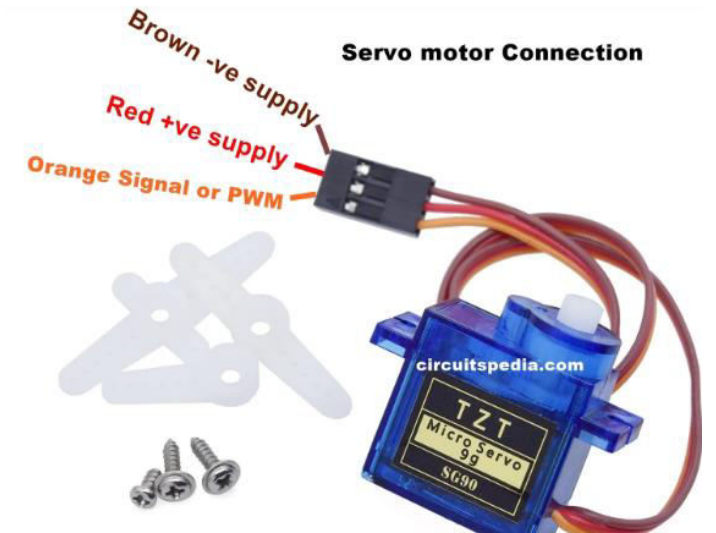
- **Force** is given for linear actuators and is often measured in Newtons (N)
- **Torque** applies to rotary actuators and is measured in Newton-meters (Nm)
- **Speed**: Specifies how fast the actuator moves, usually given in mm/s (linear) or RPM (rotary).
- **Accuracy**: How close the actuator can get to the desired position.
- **Repeatability**: The ability to return to a position consistently. High precision applications require actuators with tight tolerances for both.

Data Sheet Reading *Cont`d...*

Activity

Read the datasheet of SG90 servo motor and identify the key specifications.

Actuator Interfacing



Questions ?



Thank you !



Introduction to IoT

Department of Network and Security

Memory Types in IoT Devices



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supun.g@nsbm.ac.lk

Outline

- **Importance** of Memory in IoT Devices
- **Types** of Memories
- **Examples** of Memory Types in IoT Devices
- Memory **Consideration** for IoT Devices
- Memory **Management** Techniques
- **Arduino Uno Memory Architecture**



Importance of Memory in IoT Devices

Role of Memory in IoT Devices

1. Storing Program Code and Firmware

Non-volatile memory such as flash is used to store the program code and firmware that control the IoT device. The firmware includes the software instructions that define how the device interacts with sensors, communicates with other devices or servers, and processes data.

Example: Microcontrollers (MCUs) in IoT devices like the ESP32 or ARM Cortex-M series often have built-in flash memory where firmware and bootloaders are stored.

Role of Memory in IoT Devices *Cont`d...*

2. Data Storage and Retention

IoT devices often need to store data temporarily or permanently. This includes:

- Sensor Data
- Logs and Events
- Configuration Settings

Non-volatile memory (e.g., flash, EEPROM) is particularly important for storing data that needs to persist across power cycles, such as configuration settings.

Role of Memory in IoT Devices *Cont`d...*

3. Data Buffering and Real-Time Processing

Volatile memory like **SRAM (Static RAM)** is used for temporarily storing data that the device needs to process quickly. It serves as a workspace for the microcontroller, holding:

Buffers: Data coming in from sensors or being prepared for transmission is stored in buffers to manage timing differences between data production and processing.

Intermediate Calculations: Any computations or data manipulations performed by the device are often handled in RAM before storing results or transmitting them.

Role of Memory in IoT Devices *Cont`d...*

4. Managing Power Consumption

Memory choice significantly influences the **power consumption** of IoT devices, which is a key concern since many IoT devices run on batteries.

SRAM is typically used for low-power applications because it consumes less power when in standby mode, making it suitable for IoT sensors that spend most of their time in sleep mode.

Flash memory and **EEPROM** are also energy-efficient in retaining data without power, but write operations can consume more power, which makes it important to minimize write cycles

Role of Memory in IoT Devices *Cont`d...*

5. Facilitating Communication Protocols

Many IoT devices communicate over various protocols (like MQTT, HTTP, or CoAP) that require **temporary storage** of messages, headers, and packet data.

RAM is used for buffering incoming and outgoing data packets, managing protocol-specific states, and ensuring smooth communication with other devices or cloud servers.

Memory management is essential to avoid **buffer overflows** or data loss, which could disrupt communication and device operation.

Role of Memory in IoT Devices *Cont`d...*

6. Storing AI/ML Models for Edge Computing

Some advanced IoT devices perform **edge computing**, where data is processed locally instead of being sent to the cloud.

This involves storing machine learning models, databases, or rule sets locally, using memory like **flash** for persistent storage and **RAM** for runtime inference.

Types of Memories

I. Secondary Memory

II. Primary Memory

a) RAM

- i. SRAM
- ii. DRAM
 - i. SIMM
 - ii. DIMM
 - iii. RIMM

b) ROM

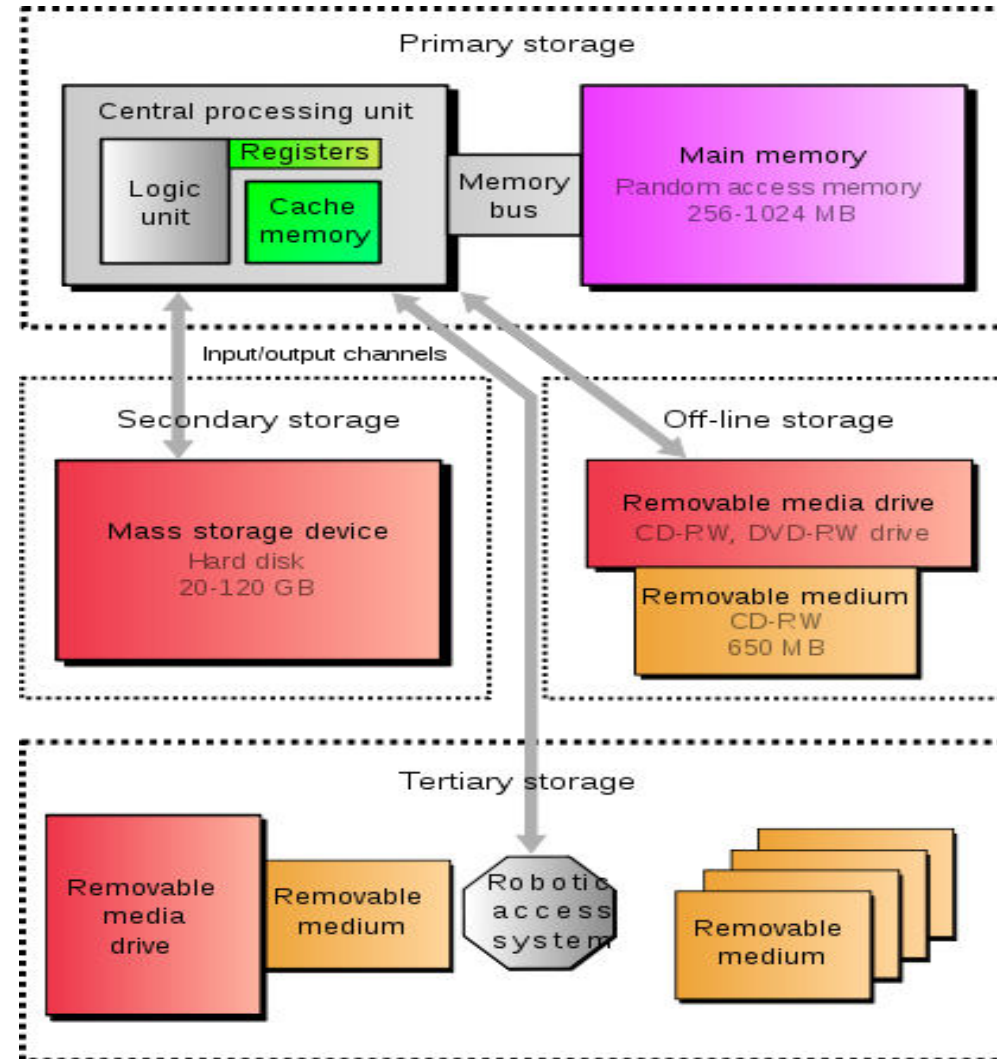
- i. PROM
- ii. EPROM

c) Hybrid

- i. EEPROM
- ii. NVRAM
- iii. Flash Memory

d) Cache Memory

e) Virtual Memory



Random Access Memory

Types of Memories – Primary Memory

RAM (Random Access Memory) Types

- Static RAM (SRAM) -1990
- Dynamic RAM (DRAM)- 1970
- Synchronous Dynamic RAM (SDRAM) -1993
- Single Data Rate Synchronous Dynamic RAM (SDR SDRAM) -1993
- Double Data Rate Synchronous Dynamic RAM (DDR SDRAM, DDR2, DDR3, DDR4) -2000
- Graphics Double Data Rate Synchronous Dynamic RAM (GDDR SDRAM, GDDR2, GDDR3, GDDR4, GDDR5)-2003

Types of Memories – RAM *Cont`d...*

SRAM (Static RAM)

- "**Static**" refers to its ability to hold stored data without needing constant refresh cycles
- A **volatile** memory
- Stores data **temporarily**
- SRAM is made up of **flip-flops** (bistable circuits) that use transistors to store a bit of data.
- Each bit in SRAM is typically stored using a **6-transistor (6T)** configuration, where six transistors form a flip-flop circuit.
- **SRAM is faster** than DRAM, but it is also more **expensive** and takes up more physical space per bit.
- **Low Power in Idle State:** Since it does not require constant refreshing, SRAM can save power when the device is idle.
- Use as CPU cache memory , program execution and storing temporary data in IoT devices

Types of Memories – RAM *Cont`d...*

SRAM Memory Cell

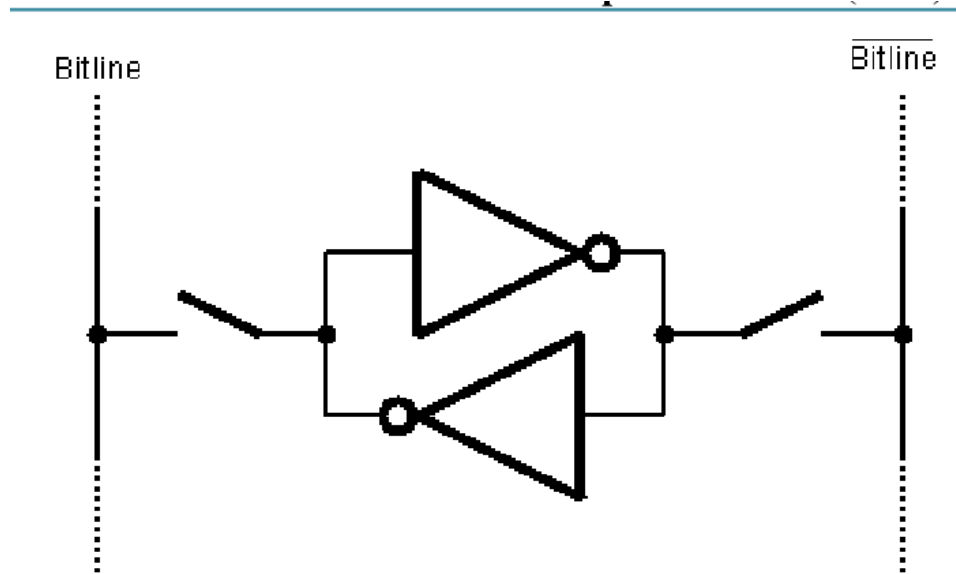


Figure 2: Basic SRAM cell

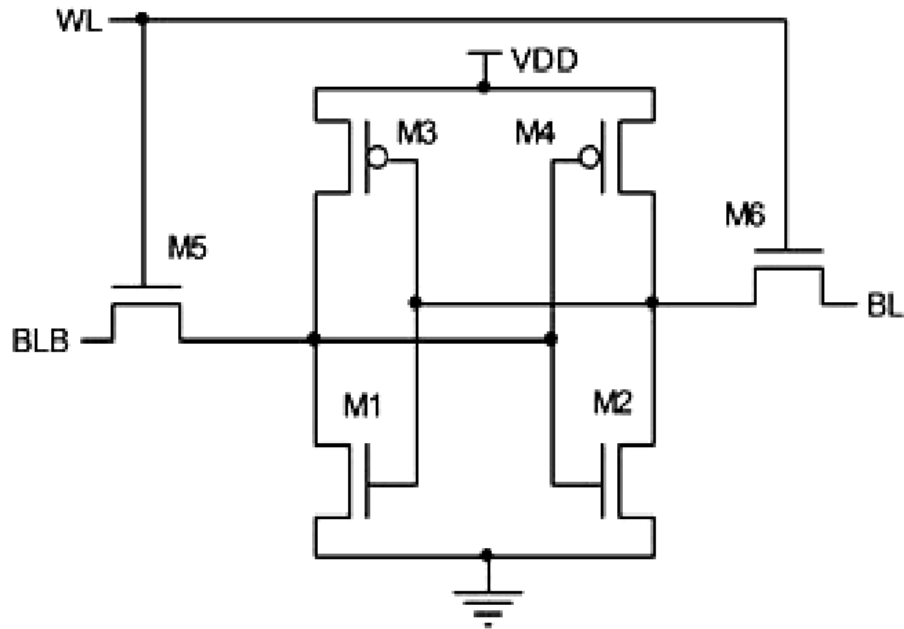


Figure 2.1.: 6T SRAM cell Structure

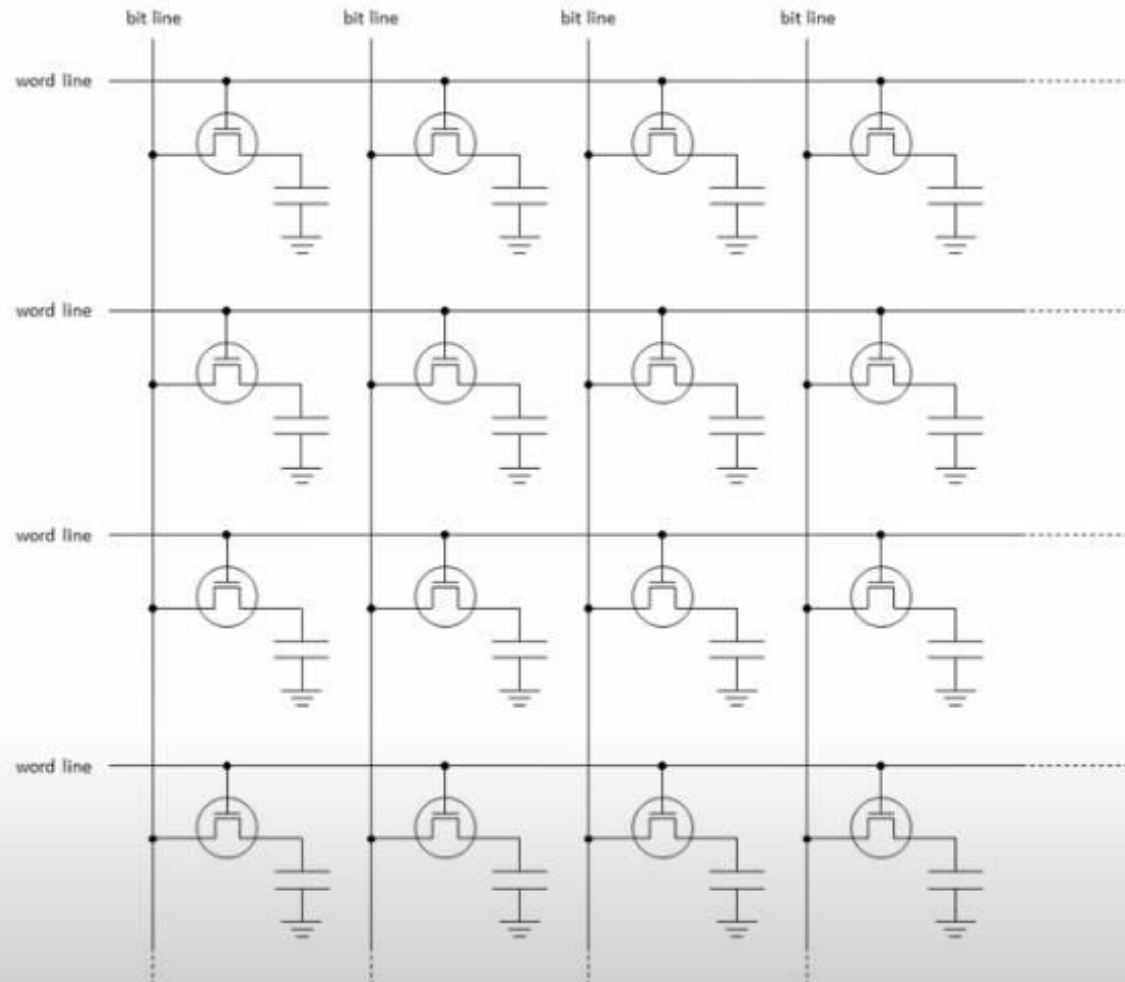
Types of Memories – RAM *Cont`d...*

DRAM (Dynamic RAM)

- The term "**dynamic**" refers to this need for continuous refresh cycles to retain data
- A **volatile** memory
- Stores data **temporarily**
- stores each bit of data in a separate **capacitor** within an integrated circuit. Because capacitors naturally lose their charge over time, DRAM requires **regular refreshing** to maintain the stored data.
- DRAM can store **a large amount of data** due to its compact cell structure, making it suitable for use in applications like main memory (e.g., RAM in computers).
- DRAM consumes **more power** than SRAM due to the need for regular refresh cycles, especially when the memory is active.

Types of Memories – RAM *Cont`d...*

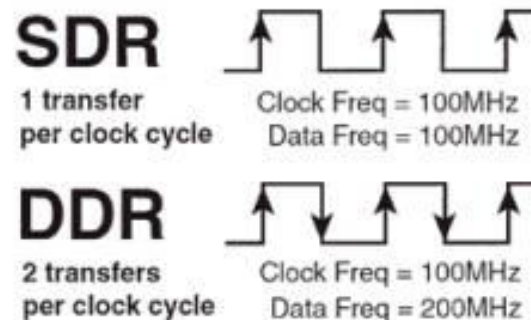
DRAM –Memory cells



Types of Memories – RAM *Cont`d...*

Types of DRAM (Dynamic RAM)

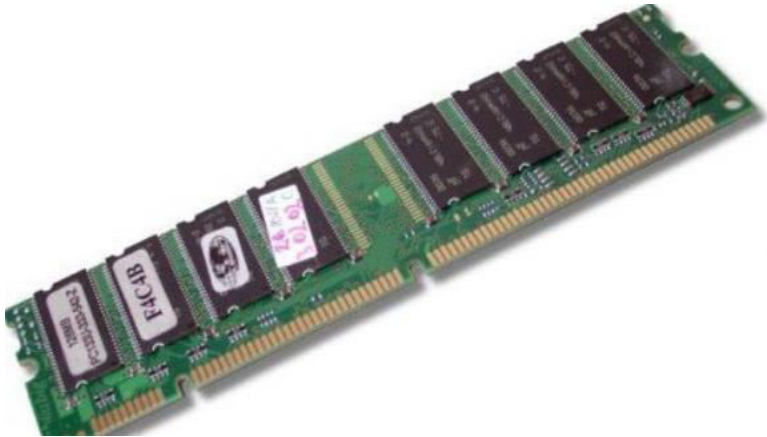
1. **SDRAM (Synchronous DRAM):** Synchronizes with the system clock, making it more efficient for modern processors. It is widely used in desktop computers and servers.
2. **DDR SDRAM (Double Data Rate SDRAM):** Transfers data on both the rising and falling edges of the clock signal, effectively doubling the data transfer rate. Common types include **DDR3, DDR4, and DDR5**, each offering higher speeds and efficiencies.



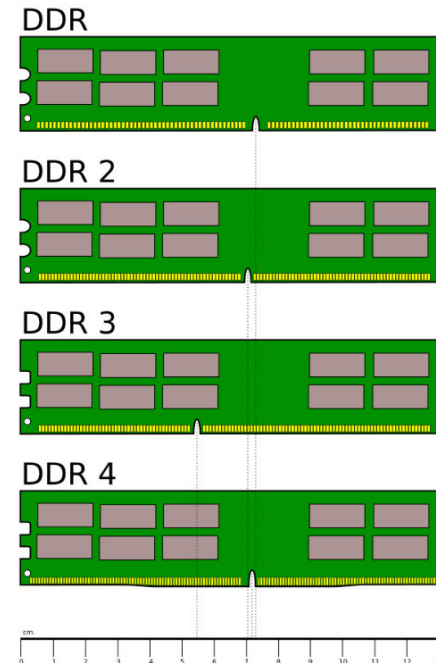
Types of Memories – RAM *Cont`d...*

Types of DRAM (Dynamic RAM)

SDRAM (Synchronous DRAM):



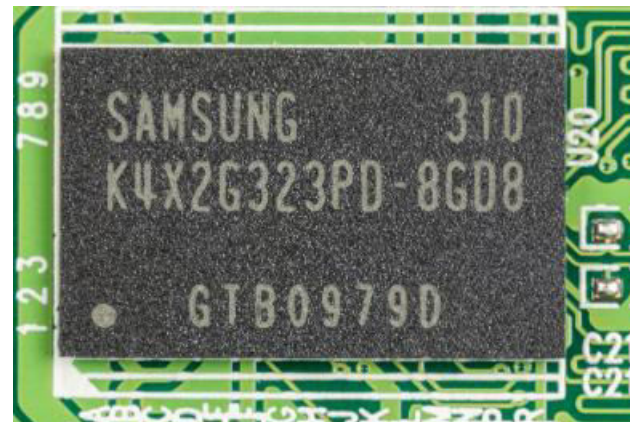
DDR SDRAM (Double Data Rate SDRAM):



Types of Memories – RAM *Cont`d...*

Types of DRAM (Dynamic RAM)

3. **LPDDR (Low Power DDR):** A variant designed for mobile devices, offering reduced power consumption, making it suitable for IoT devices, smartphones, and tablets.



Types of Memories – RAM *Cont`d...*

Types of DRAM (Dynamic RAM)

4. GDDR SDRAM (Graphics DDR SD RAM)

A type of memory specifically designed for use in **graphics processing units (GPUs)** and **video cards**. It is optimized for the **high bandwidth** demands of **graphics rendering**, **3D gaming**, **video processing**, and **AI workloads**. GDDR SDRAM shares some architectural similarities with **DDR SDRAM** (used in regular system memory) but has been adapted to better suit the unique needs of graphics hardware.



Types of Memories – RAM Cont'd...

From Factors of DRAM (Dynamic RAM)

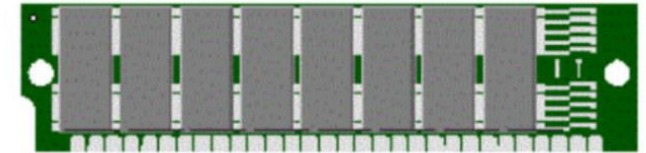
1. SIMM (Single Inline Memory Modules)

SIMMs were used in early personal computers and **workstations** before being replaced by **DIMMs** as systems transitioned to **64-bit architectures**.

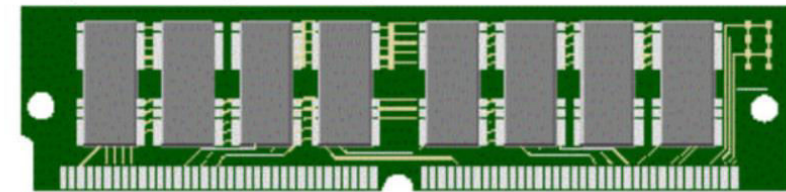
SIMMs were among the earliest memory modules and were commonly used in computers from the **1980s to the mid-1990s**.

Required to be installed in **pairs** on systems with **64-bit data buses** (like early **Pentium** systems), as a single SIMM only provided **32-bit** data.

30-pin SIMM (3.5 x .75")



72-pin SIMM (4.25 x 1")



Types of Memories – RAM *Cont`d...*

From Factors of DRAM (Dynamic RAM)

2. DIMM (Dual Inline Memory Module)

Usage: Desktops, servers, and high-end workstations.

DIMMs have **separate electrical contacts** on each side of the module (dual inline), allowing for more connections and thus more capacity and bandwidth.

Non-ECC DIMMs are common in consumer-grade desktops, while **ECC (Error-Correcting Code)** DIMMs are used in servers and workstations to improve data integrity.



Types of Memories – RAM *Cont`d...*

From Factors of DRAM (Dynamic RAM)

2.1 SO-DIMM (Small Outline DIMM)

Usage: Laptops, compact desktops, embedded systems, and some IoT devices.

SO-DIMMs are designed for devices where **space is limited**, making them ideal for portable devices like laptops.

Lower power consumption versions like **LPDDR** (Low Power DDR) are often used to further reduce energy use in portable devices.



Types of Memories – RAM Cont'd...

From Factors of DRAM (Dynamic RAM)

3. RIMMs (Rambus Inline Memory Modules)

RIMMs were developed by **Rambus Inc.** as a high-performance alternative to DIMMs and were used primarily in the late **1990s** and early **2000s**.

They used **RDRAM (Rambus DRAM)** technology, which offered **higher data transfer rates** compared to **SDRAM** at the time.

RIMMs were briefly popular in **high-end PCs** and **workstations**, including **early Intel Pentium 4 systems**, but were quickly outpaced by the **DDR SDRAM** family due to cost and performance trade-offs.



Types of Memories – RAM *Cont`d...*

ECC RAM vs Non ECC RAM

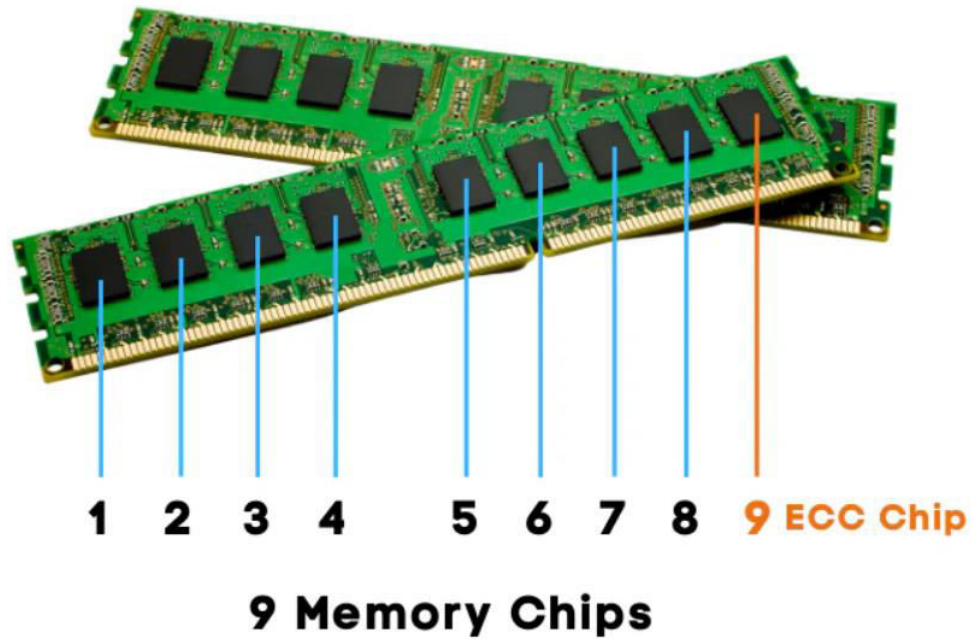
ECC RAM (Error-Correcting Code RAM) is a type of computer memory that can **detect** and **correct** common types of **data corruption**. It is typically used in **servers, workstations**, and other **mission-critical** systems where **data integrity** is crucial.

Non-ECC RAM is the standard type of memory used in most **consumer-grade** computers, such as **desktop PCs, laptops**, and **gaming systems**. It does not have the capability to **detect** or **correct** errors in memory data.

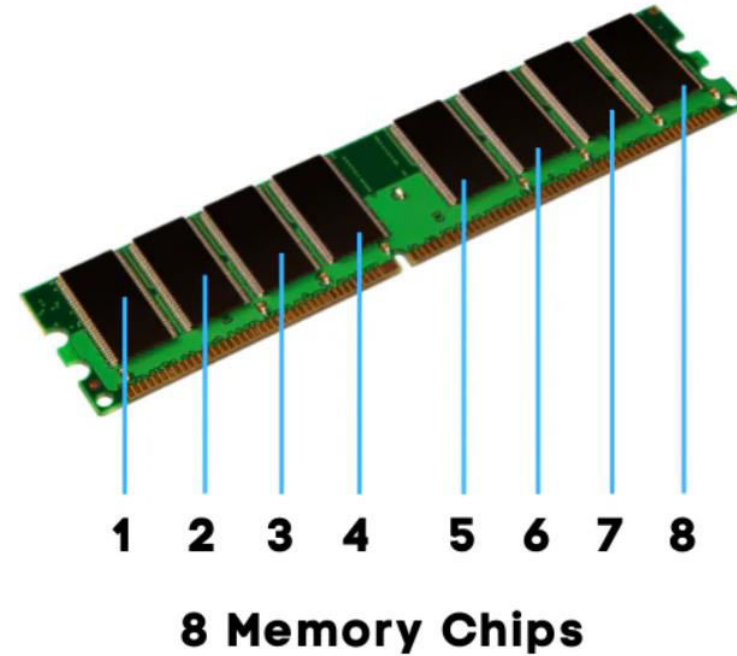
Types of Memories – RAM *Cont`d...*

ECC RAM vs Non ECC RAM

ECC Memory



Non-ECC Memory



Types of Memories – RAM *Cont`d...*

Applications of DRAM (Dynamic RAM)

- ✓ **Main Memory in Computers and Servers:** DRAM serves as the primary system memory in PCs, laptops, and servers. It holds the operating system, application data, and running programs, enabling fast access and processing by the CPU.
- ✓ **Embedded Systems and IoT Devices:** For devices that need to process and store a significant amount of data locally, such as IoT gateways or edge computing devices, DRAM can be used to handle data buffers and processing tasks.
- ✓ **Graphics and Video Applications:** Variants of DRAM like **GDDR (Graphics DDR)** are used as video RAM (VRAM) in graphics cards, supporting fast rendering of images and videos.

Examples of Memory Types in IoT Devices

Examples of Memories in IoT Devices

Specifications	ESP8266	ESP32
MCU	Xtensa® Single-Core 32-bit L106	Xtensa® Dual-Core 32-bit LX6 600 DMIPS
802.11 b/g/n Wi-Fi	Yes, HT20	Yes, HT40
Bluetooth	None	Bluetooth 4.2 and below
Typical Frequency	80 MHz	160 MHz
SRAM	160 kBytes	512 kBytes
Flash	SPI Flash , up to 16 MBytes	SPI Flash , up to 16 MBytes
GPIO	17	36
Hardware / Software PWM	None / 8 Channels	1 / 16 Channels
SPI / I2C / I2S / UART	2/1/2/2	4/2/2/2
ADC	10-bit	12-bit
CAN	None	1
Ethernet MAC Interface	None	1
Touch Sensor	None	Yes
Temperature Sensor	None	Yes
Working Temperature	- 40°C ~ 125°C	- 40°C ~ 125°C

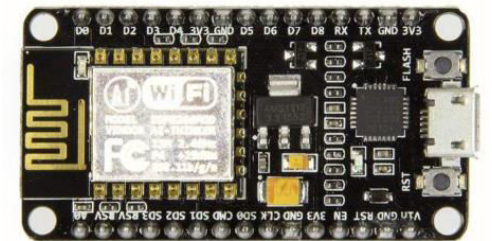


Figure: ESP8266 vs ESP32

Read Only Memory

Types of Memories – ROM

1. ROM (Read-Only Memory)

ROM is a **non-volatile memory** that stores **permanent data** used by a computer or device.

Data in ROM cannot be modified (read-only) during normal operation.

Commonly used in **BIOS** (Basic Input/Output System) in computers, and in **microcontrollers** for storing **boot-up instructions**.

Types of Memories – ROM *Cont`d...*

2. PROM (Programmable Read-Only Memory)

PROM is a type of **ROM** that can be **programmed** once after manufacturing.

Non-volatile like other ROM types.

Programming can only be done once using a special **PROM programmer**.

Once written, data cannot be changed or erased.

An example of a PROM is a computer's **BIOS** (basic input/output system) in early computers. Today, PROM in computers has been replaced by EEPROM (electrically erasable programmable read-only memory).

Types of Memories – ROM *Cont`d...*

3. EPROM (Erasable Programmable Read-Only Memory)

EPROM is a type of PROM that can be **erased** and **reprogrammed**.

Erasable using **ultraviolet (UV) light** through a special window in the chip.

Reprogramming requires a **special device** called an **EPROM programmer**.

Non-volatile, but allows for data updates.

Used in **development** environments where firmware or data might change, such as in **embedded systems** before moving to a final design.

Types of Memories – ROM *Cont`d...*

4. Hybrid Memory

Hybrid memory combines the features of **volatile** and **non-volatile memory**, aiming for a balance between **speed** and **data retention**.

Can refer to memory systems that integrate **NAND flash** with **DRAM**, providing **fast access speeds** with **non-volatile** backup.

Hybrid drives in computers (combining **SSD** and **HDD**), **memory systems** in **IoT devices** that require quick access with backup storage.

Types of Memories – ROM *Cont`d...*

5. EEPROM (Electrically Erasable Programmable Read-Only Memory)

EEPROM is a type of **ROM** that can be **electrically erased** and **reprogrammed**.

Can be reprogrammed **electrically** without needing to remove the chip from the circuit.

Slower than **Flash Memory** but suitable for storing small amounts of data like **device configurations**.

Types of Memories – ROM *Cont`d...*

6. NVRAM (Non-Volatile Random Access Memory)

NVRAM is a type of **memory** that combines **fast access** like **RAM** with **non-volatility**.

Faster access compared to EEPROM and Flash.

Typically has a **limited write endurance** (number of times data can be written before it wears out).

Types of Memories – ROM *Cont`d...*

7. Flash Memory

Flash memory is a **non-volatile memory** that can be **electrically erased** and **reprogrammed in blocks**.

Much **faster** than **EEPROM** for large data blocks.

Comes in two main types: **NAND** and **NOR** flash, with **NAND** being common in **SSDs** and **USB drives** and **NOR** used in **firmware storage**.

Write endurance is typically higher than EEPROM, making it suitable for **storage devices**.

Cache Memory

Types of Memories – Cache

Cache Memory

Cache memory is a **high-speed memory** located close to the **CPU** that stores **frequently accessed data**.

Acts as a **buffer** between the CPU and **main memory (RAM)**, reducing **latency**.

Comes in different levels (L1, L2, L3), with **L1** being the fastest and smallest, and **L3** being larger but slower.

Volatile, so data is lost when the power is off.

Virtual Memory

Types of Memories – Virtual

Virtual Memory

Virtual memory is a **memory management technique** that uses **part of the storage drive** as if it were **additional RAM**.

Allows the system to **handle larger applications** than the physical RAM would allow.

Slower than **physical RAM** because accessing data from a **hard drive** or **SSD** takes more time.

Operating systems use virtual memory to handle **memory-intensive applications**, preventing **out-of-memory errors** by utilizing storage as an overflow space.

Memory Consideration for IoT Devices

Memory Considerations for IoT Devices

1. Power Consumption

IoT devices often operate on batteries or have limited power sources, making power consumption a critical factor.

SRAM consumes power continuously but is fast, while **DRAM** has refresh cycles that can add to power use.

Flash memory and **EEPROM** consume less power during standby but can draw significant power during write operations. Thus, write cycles should be minimized.



Memory Considerations for IoT Devices *Cont`d...*

2. Memory Size and Cost

IoT devices often need to balance the amount of RAM and non-volatile memory due to cost constraints.

Devices with more complex functions (e.g., edge computing, local data analysis) may require more RAM and storage space.



Memory Considerations for IoT Devices *Cont`d...*

3. Data Retention and Longevity

Flash memory and **EEPROM** have a limited number of write/erase cycles, which can be a concern for IoT devices that frequently update data.

Understanding the **wear leveling** and **data retention** characteristics of non-volatile memory is crucial for ensuring long-term reliability.



Memory Considerations for IoT Devices *Cont`d...*

4. Security

Sensitive data like encryption keys or user credentials often need to be stored securely.

Some microcontrollers include **secure memory regions** or **hardware security modules (HSMs)** to protect critical information from unauthorized access.



Memory Management Techniques

Memory Management Techniques

Managing memory effectively is important for ensuring the smooth operation of IoT devices, especially given their constrained resources.

Data Compression: Reduces the size of data before storing it in flash or transmitting it over a network.

Memory Optimization: Choosing data structures that minimize memory usage (e.g., using `uint8_t` instead of `int` when values are small)

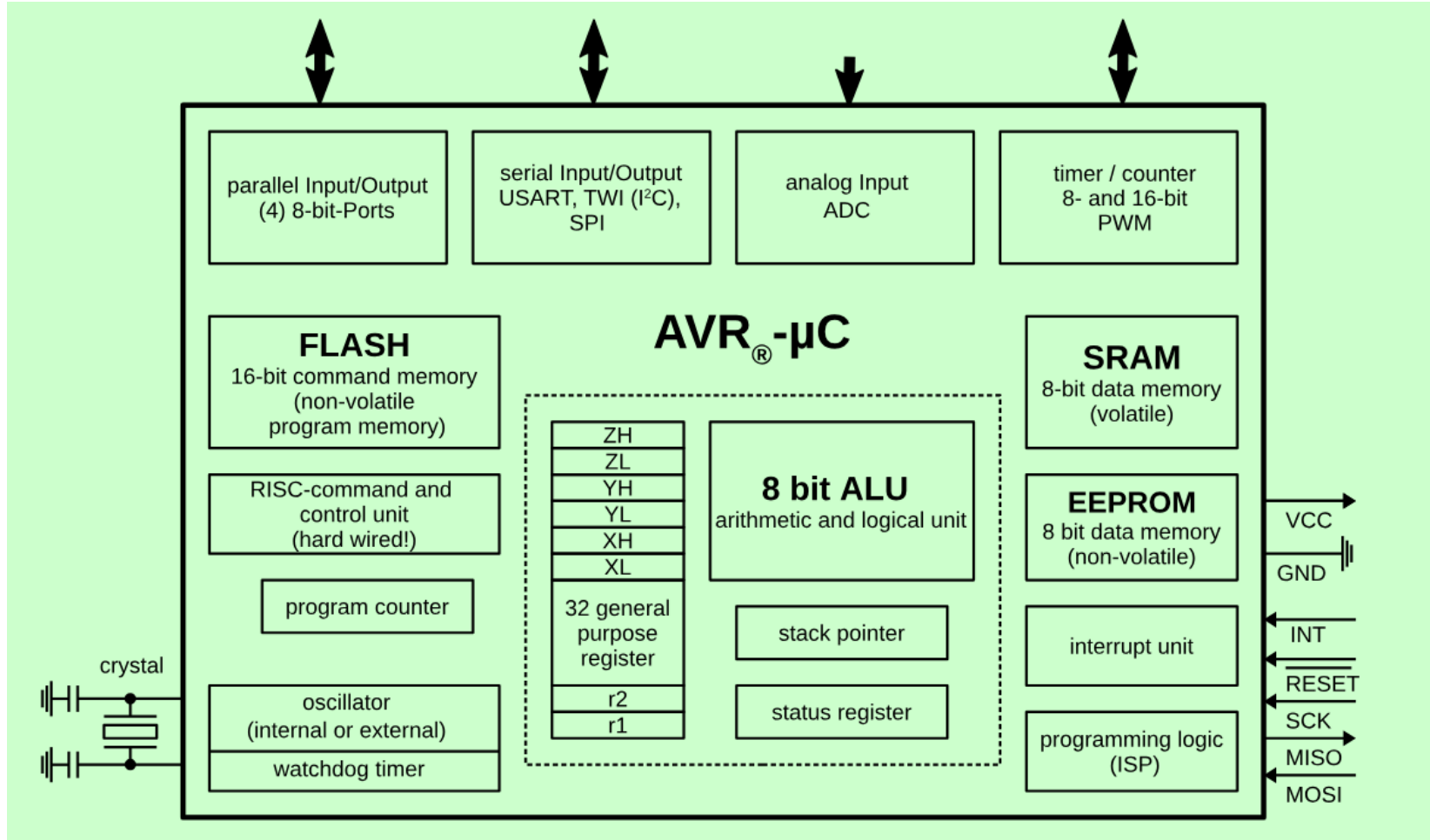
Garbage Collection: Especially in higher-level IoT platforms, managing how memory is allocated and freed can prevent leaks.

Memory Management Techniques *Cont`d...*

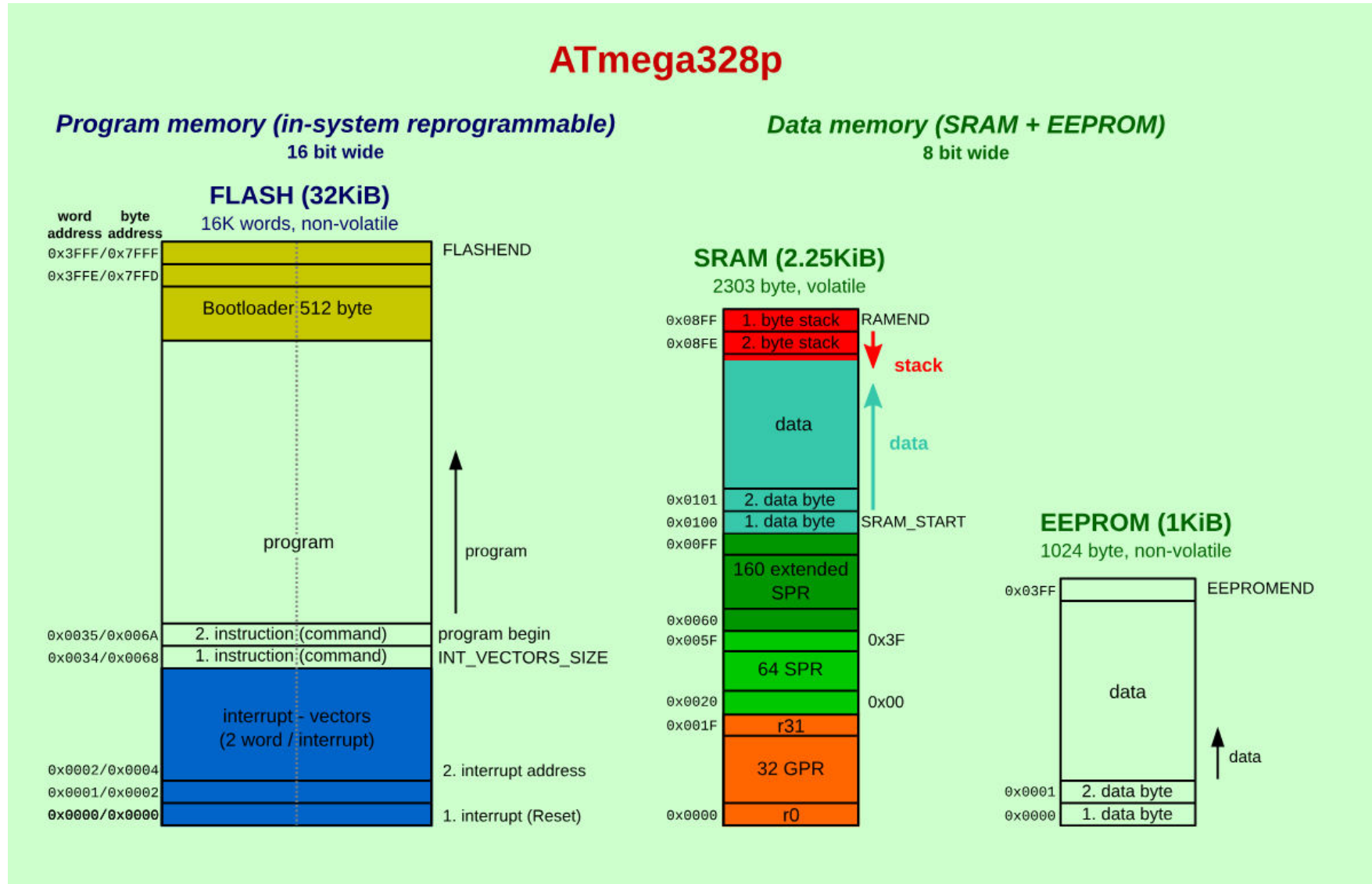
Dynamic Memory Allocation: Allocating memory during the runtime of an application. Allows IoT devices to adjust memory usage based on current needs, making efficient use of the limited RAM. Suitable for applications where memory requirements change frequently.

Virtual Memory Techniques: Extending the physical memory by using storage space (e.g., a part of the flash memory) as additional RAM. Allows large applications or data sets to run on devices with limited RAM.

Arduino Uno Memory Architecture



Arduino Uno Memory Architecture *Cont`d...*



Questions ?



Thank you !



Internet of Things (IoT)

Department of Computer Security and Network Systems

Programming and Data collection



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Connecting Sensors to the Arduino Board

Analog Sensors

Analog sensors, like potentiometers and temperature sensors, provide a range of values.

1. Connect the sensor's output to an analog pin on the Arduino board.
2. Use the ``analogRead()`` function to read the sensor's value.

Digital Sensors

Digital sensors, like push buttons and motion sensors, output either high or low signals.

1. Connect the sensor's output to a digital pin on the Arduino board.
2. Use the ``digitalRead()`` function to read the sensor's state.

I2C and SPI Sensors

These communication protocols enable you to connect multiple sensors to the Arduino board using fewer pins.

1. Connect the sensor's SDA, SCL, and other required pins to the corresponding Arduino pins.
2. Use libraries specific to the I2C or SPI protocol for communication.

Reading Sensor Data

1

Analog Sensor Data

Use the ``analogRead()`` function to read the sensor's value, which is a number between 0 and 1023.

3

I2C and SPI Data

Use libraries to communicate with the sensor and retrieve data according to the protocol's specific commands.

2

Digital Sensor Data

Use the ``digitalRead()`` function to read the sensor's state, which will be either HIGH (1) or LOW (0).

4

Conversion and Scaling

Often, you need to convert the raw sensor data to a meaningful value, such as temperature or distance, using a formula or lookup table.

Processing and Analyzing Sensor Data

1

Data Cleaning

Remove any erroneous or outlier data points to ensure accurate analysis.

2

Filtering

Apply filters to remove noise or smooth data for clearer insights.

3

Calculations and Comparisons

Perform calculations to derive new information from the data, such as averages, differences, or trends.

4

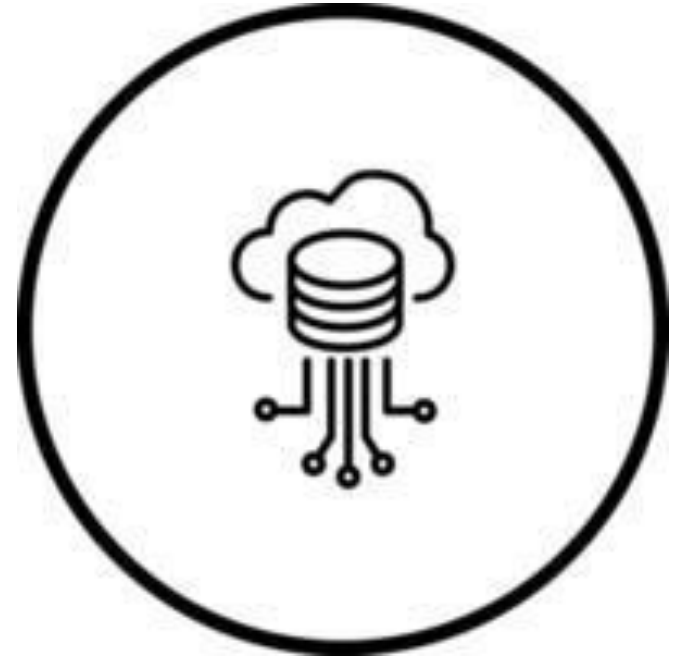
Decision Making

Use the processed data to make decisions based on predefined criteria or thresholds.



Storing and Retrieving Data

Storage Options	Description
SD Card	Store data persistently on an SD card for later retrieval.
EEPROM	Store small amounts of data directly on the Arduino board, ideal for settings or calibration values.
Cloud Databases	Upload data to cloud services for remote access and analysis.



Visualizing Sensor Data

Serial Monitor

The Arduino IDE's built-in Serial Monitor allows you to view real-time data in a simple text format.

Graphical Libraries

Libraries like Processing or Plotter enable you to create visually appealing charts and graphs.

Web Interfaces

Create web interfaces using technologies like Node.js or PHP to display data and control the Arduino remotely.

Practical Applications and Project Ideas



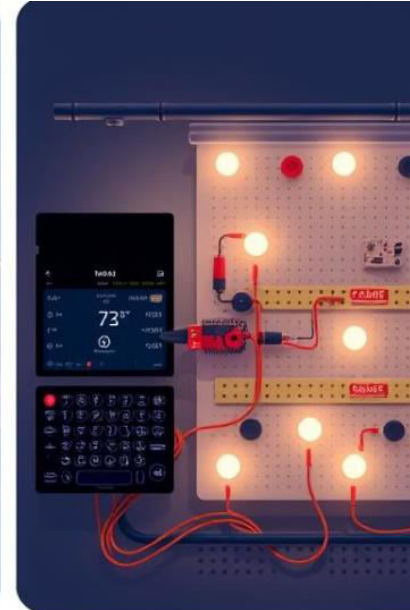
Weather Station

Measure temperature, humidity, and pressure to track local weather conditions.



Smart Garden

Monitor soil moisture and temperature to automate plant watering and lighting.



Automated Lighting System

Use light sensors to control lighting based on ambient light levels.



Robotic Arm

Build a robotic arm with Arduino control to perform tasks like picking and placing objects.





Thank You

Internet of Things (IoT)

Department of Computer Security and Network Systems

Introduction to IoT Hardware



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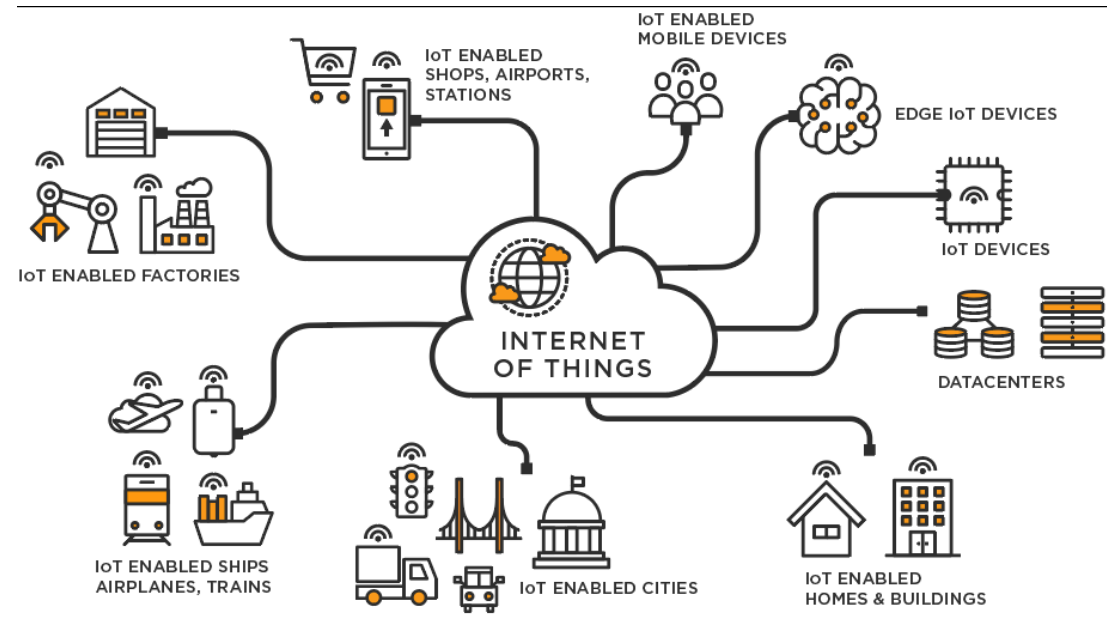


What is IoT ?



IoT ?

The Internet of Things (IoT) describes the network of **physical objects** that are **embedded with sensors, software, and other technologies** for the purpose of **connecting and exchanging data** with other **devices and systems over the internet**.

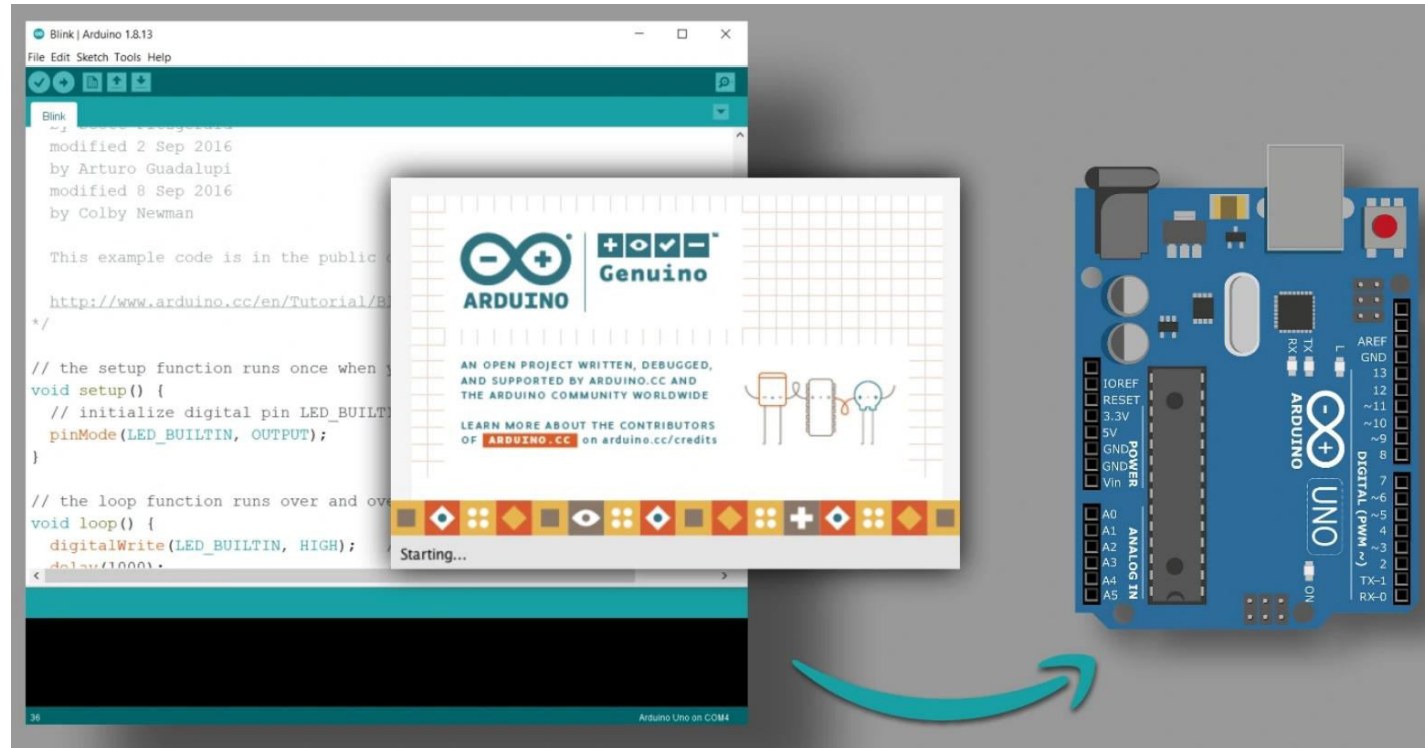


Then what is Arduino ?

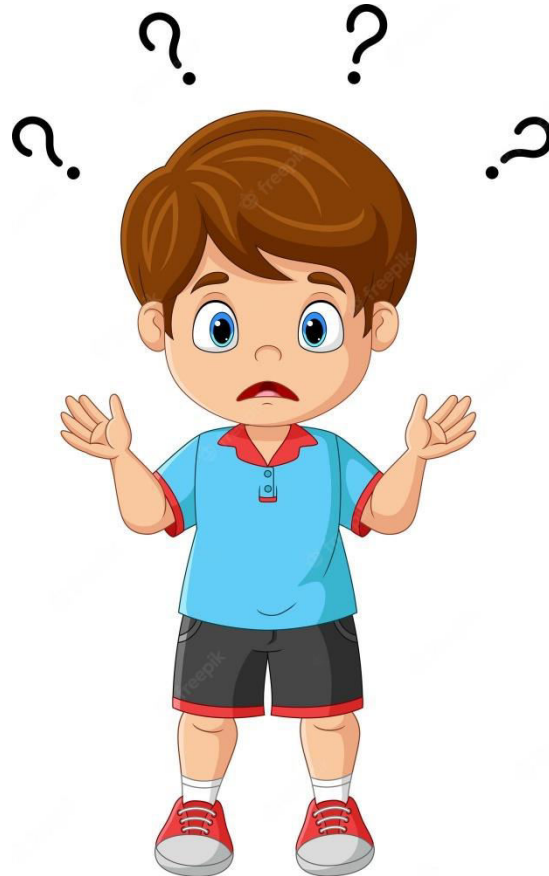


Arduino ?

Arduino is an **open-source electronics platform based on easy-to-use hardware and software.**



Do we have any alternatives for Arduino?



Do we ?

- Raspberry Pi
- Teensy 3.6
- Launchpad MSP430



Top 10 Strategic IoT Technologies and Trends - GARTNER

- 1) **Trend No. 1: Artificial Intelligence (AI):** “Data is the fuel that powers the IoT and the organization’s ability to derive meaning from it will define their long term success.”
- 2) **Trend No. 2: Social, Legal and Ethical IoT:** These include ownership of data and the deductions made from it, algorithmic bias, privacy and compliance with regulations such as the General Data Protection Regulation. “Successful deployment of an IoT solution demands that it’s not just technically effective but also socially acceptable.”
- 3) **Trend No. 3: Infonomics and Data Broking:** The theory of infonomics takes monetization of data further by seeing it as a strategic business asset to be recorded in the company accounts. By 2023, the buying and selling of IoT data will become an essential part of many IoT systems.

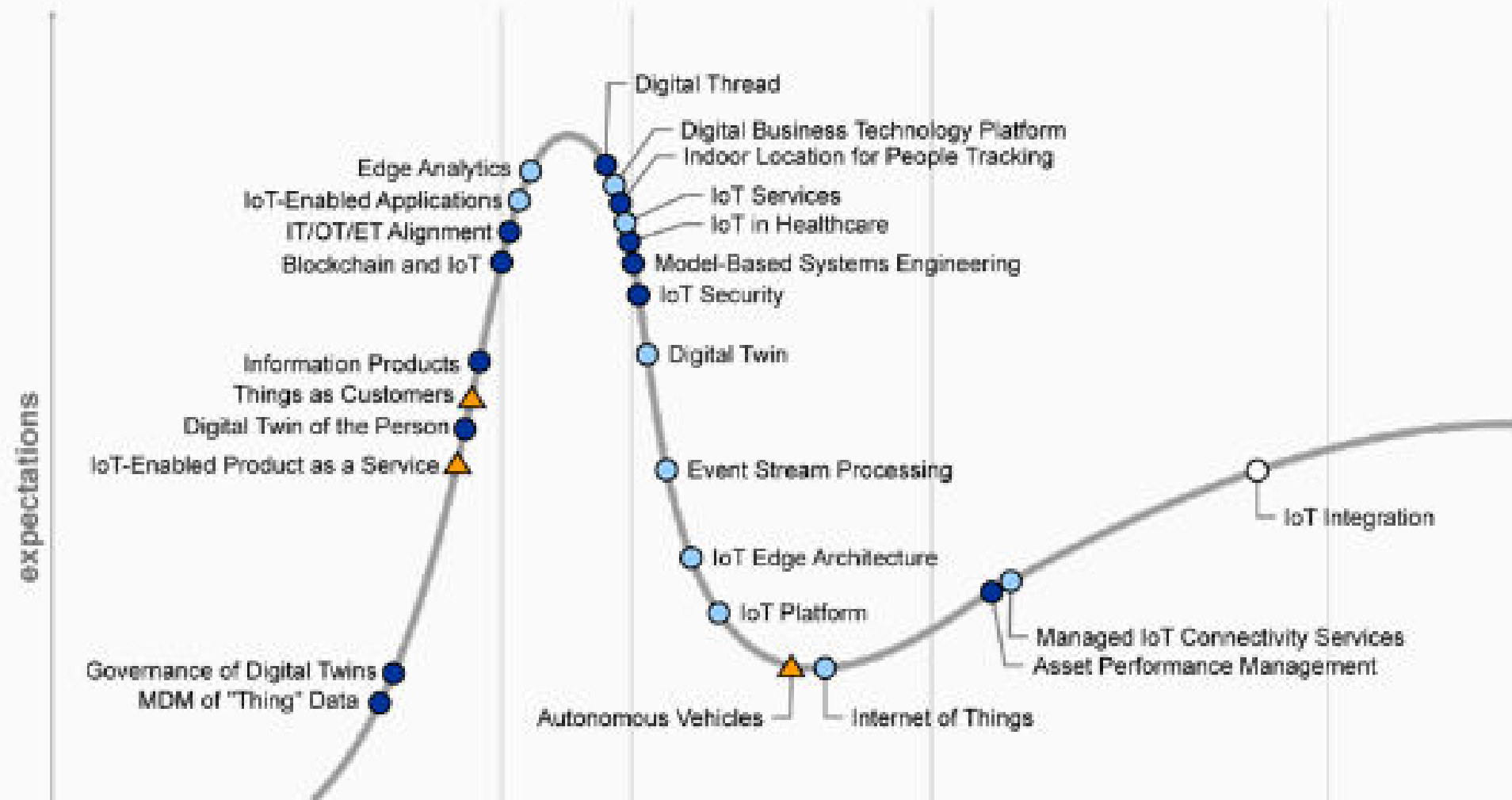
Top 10 Strategic IoT Technologies and Trends – GARTNER (cont'd)

- 4) **Trend No. 4: The Shift from Intelligent Edge to Intelligent Mesh:** The shift from centralized and cloud to edge architectures is well under way in the IoT space. These mesh architectures will enable more flexible, intelligent and responsive IoT systems — although often at the cost of additional complexities.
- 5) **Trend No. 5: IoT Governance:** As the IoT continues to expand, the need for a governance framework that ensures appropriate behaviour in the creation, storage, use and deletion of information related to IoT projects will become increasingly important.
- 6) **Trend No. 6: Sensor Innovation:** The sensor market will evolve continuously through 2023. New sensors will enable a wider range of situations and events to be detected.

Top 10 Strategic IoT Technologies and Trends – GARTNER (cont'd)

- 7) **Trend No. 7: Trusted Hardware and Operating System:** ‘.. by 2023, we expect to see the deployment of hardware and software combinations that together create more trustworthy and secure IoT systems...’.
- 8) **Trend 8: Novel IoT User Experiences:** User experience driven by 4 factors: new sensors, new algorithms, new experience architectures and context, and socially aware experiences.
- 9) **Trend No. 9: Silicon Chip Innovation:** By 2023, it’s expected that new special-purpose chips will reduce the power consumption required to run IoT devices.
- 10) **Trend No. 10: New Wireless Networking Technologies for IoT:** IoT networking involves balancing a set of competing requirements. In particular they should explore 5G, the forthcoming generation of low earth orbit satellites, and backscatter networks.

Hype Cycle for the Internet of Things, 2020



As of July 2020

On the Rise

- MDM(Master Data Management) of “Thing” Data
- Governance of Digital Twins
- IoT-Enabled Product as a Service
- Digital Twin of the Person
- Things as Customers
- Information Products
- Block chain and IoT

At the Peak

- IT/OT/ET (information/operation/enterprise technology)Alignment
- IoT-Enabled Applications
- Edge Analytics
- Digital Thread
- Digital Business Technology Platform
- Indoor Location for People Tracking
- IoT in Healthcare
- IoT Services
- Model-Based Systems Engineering

Sliding Into the Trough

- IoT Security
- Digital Twin
- Event Stream Processing
- IoT Edge Architecture
- IoT Platform
- Autonomous Vehicles
- Internet of Things

Climbing the Slope (closing the Plato)

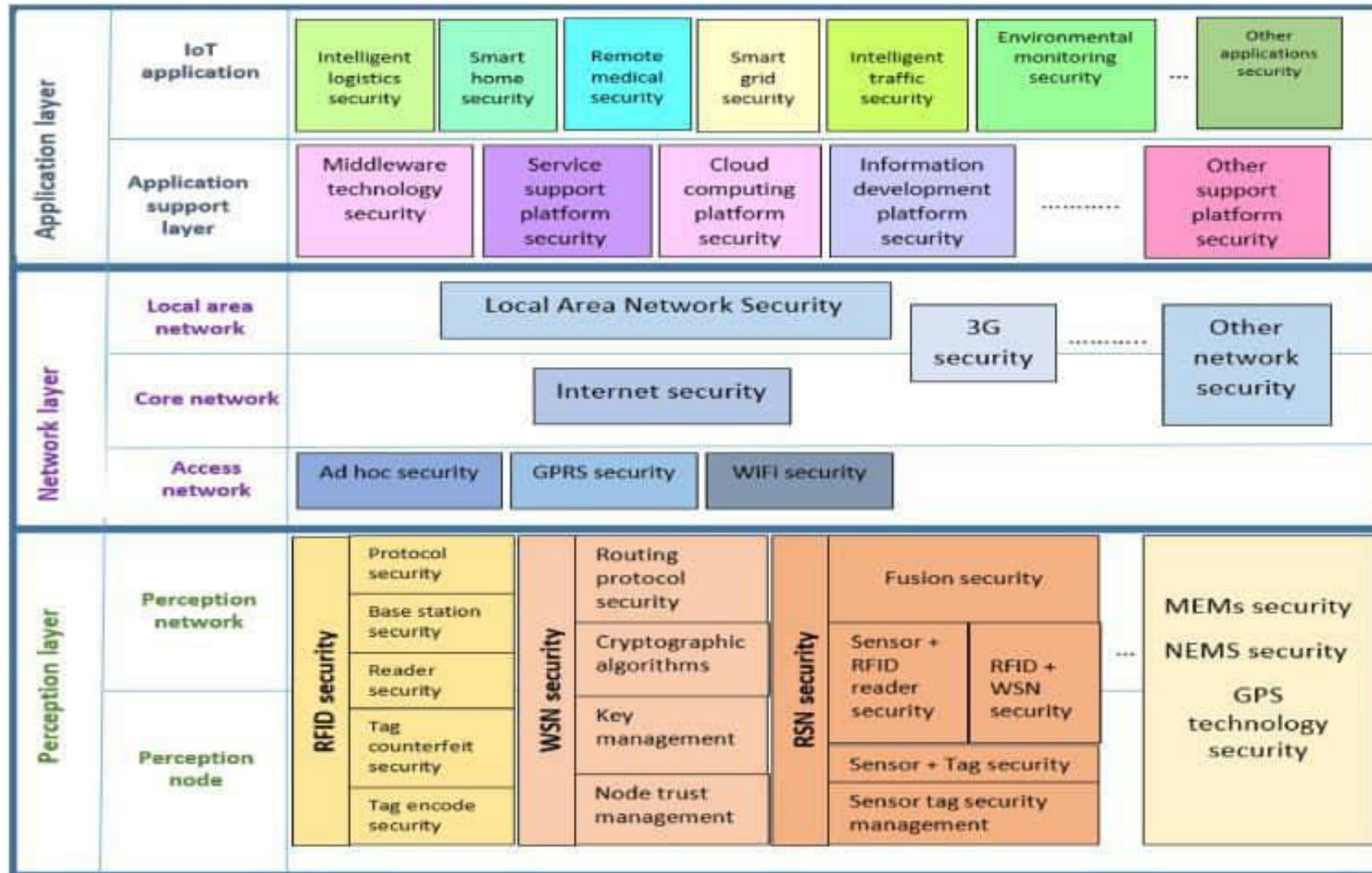
- Asset Performance Management
- Managed IoT Connectivity Services
- IoT Integration

IOT CHALLENGES

Security, privacy and sharing of Data

- Because IoT devices are closely connected, compromise of one may lead to grater risk .The manufacturers don't update their devices regularly/ or at all, leave them vulnerable to cybercriminals.
- Privacy is another major concern due to security vulnerability.
- Lack of standardization lead to many problems.
- Challenges with IIoT:
 - i. Security of data
 - ii. Reliability and stability of IIoT sensors and boards.
 - iii. Connectivity of all the systems in IIoT setup
 - iv. Blending legacy systems – IIoT is new in the market

Summary of Opportunities



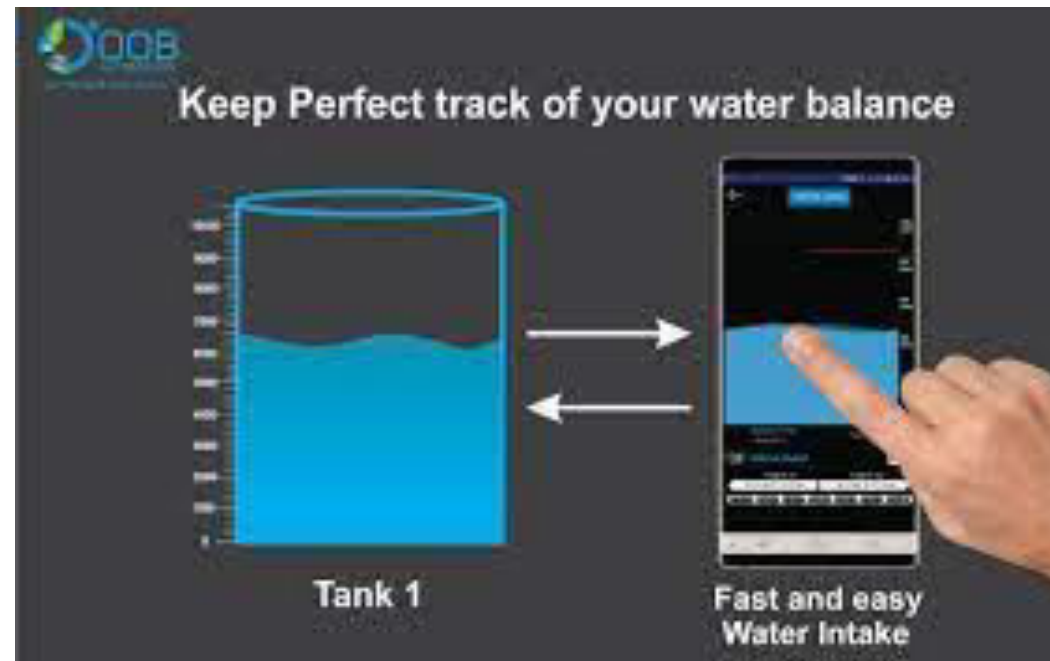
How do you select a project ?



Practical Scenario

Example :- **Water level measurement application for the university**

What is your approach ?



What is your approach ?

What are the devices that you are going to select ?



How to select the device ?

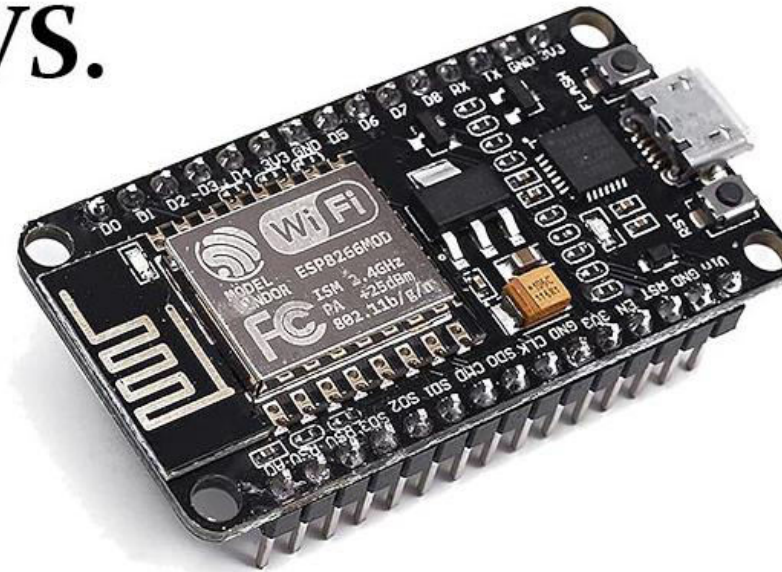
Device List ?

1. Arduino Uno / Node MCU (Esp32) Board
2. Ultrasonic Sensor
3. GSM Module

How to select the device ?



VS.



How to select the device ?



	
Price Points	\$19.99-\$23.00
Dimension	2.7 in x 2.1 in
Processor	Atmega328P
Clock Speed	16MHz
Flash Memory (kB)	32
EEPROM (kB)	1
SRAM (kB)	2
Voltage Level	5V
Digital I/O Pins	14
Digital I/O with PWM Pins	6
Analog Pins	6
USB Connectivity	Standard A/B USB
Shield Compatibility	Yes
Ethernet/Wi-Fi/Bluetooth	No (a Shield/module can enable it)

Parameters	Specification
Microcontroller	ESP8266
Memory	32 bit
Processor	TenSilica L 106
Processor Clock	80MHz-160MHz
RAM	36Kb
Storage	16 Mb
Built-in WiFi	2.4GHz supports 802.11 b/g/n
ADC Pin	1(10bit Resolution)
GPIO pins	10
Operating Voltage	3.0V 3.6V
Operating Current	80mA(Average)
Operating Temperature Range	-40oC - 125 oC

How to select the device ?



How to select the device ?



Electrical Parameters	HC-SR04 Ultrasonic Module
Operating Voltage	DC-5V
Operating Current	15mA
Operating Frequency	40KHZ
Farthest Range	4m
Nearest Range	2cm
Measuring Angle	15 Degree
Input Trigger Signal	10us TTL pulse
Output Echo Signal	Output TTL level signal, proportional with range
Dimensions	45*20*15mm

	Pulse width output / Serial Output
Operating Voltage	DC 3.0-5.5V
Working current	Less than 8mA
Probe frequency	40KHz
Farthest range	600cm
Recent range	20cm
Distance accuracy	+/- 1cm
Resolution	1mm
Measuring angle	75 degree
Enter the trigger signal	1, 10uS above the TTL pulse 2, the serial port to send instructions 0X55
Output the echo signal	Output pulse width level signal, or TTL
Wiring	3-5.5V (power positive) Trig (RX) RX Echo (output) TX GND (power supply negative)
Product Size	L42 * W29 * H12 mm
Operating temperature	-20 ° C to + 70 ° C
Product color	PCB board is blue

How to select the device ?



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Thank You

Introduction to IoT

Department of Computer Security and Network Systems

Introduction



Supun Gajendrasinghe

supun.g@nsbm.ac.lk

Recommended Readings



- IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017
- Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015
- Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill HigherEducation

Outline

- What is IoT ?
- Your Roles in IoT Projects
- History
- Genesis of IoT
- IoT Applications
- Your Project Ideas



What is IoT ?

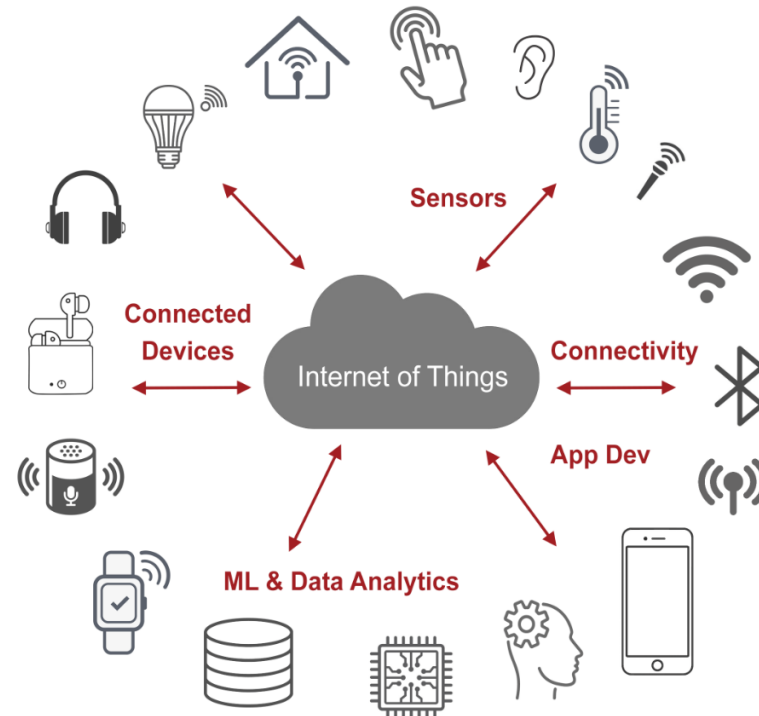
The term "**Internet of Things**" (**IoT**) is made up of two key concepts: "**Internet**" and "**Things**".

- The **Internet** is the global network that connects millions of computers and devices, allowing them to communicate with each other. When devices are connected to the Internet, they can share information and be controlled remotely.
- **Things** refer to physical objects that are equipped with technology such as sensors, software, and other components that enable them to collect and exchange data over the Internet.

What is IoT ? *Cont`d...*

Putting It Together:

The "**Internet of Things**" means connecting various physical objects ("things") to the Internet, so they can communicate, share data, and work together to perform tasks automatically or provide useful information.



What is IoT ? *Cont`d...*

Goal of IoT:

Connecting the Unconnected.

Objects that are not currently joined to a computer network-Internet, will be connected so that they can communicate and interact with people and other objects.



Your Roles in an IoT Project



What Computing
People do in IoT
projects ?

Your Roles in an IoT Project *Cont`d...*

Embedded Software Developer:

Role: Develop firmware and software for IoT devices.

Students with skills in C/C++, Python, and knowledge of embedded systems can contribute significantly in this role.



Your Roles in an IoT Project *Cont`d...*

Cloud and Backend Developer:

Role: Design and implement cloud services and backend infrastructure for IoT applications.

Students with experience in cloud platforms (e.g., AWS, Azure, Google Cloud), database management, and server-side programming languages like Node.js, Java, or Python are well-suited for this role.



Your Roles in an IoT Project *Cont`d...*

Data Analyst/Data Scientist:

Role: Analyze data generated by IoT devices to extract meaningful insights.

Students with a background in data science, machine learning, and tools like R, Python, and MATLAB can excel in this role.



Your Roles in an IoT Project *Cont`d...*

Network and Communication Engineer:

Role: Implement and manage the communication infrastructure for IoT systems.

Students with knowledge in networking, IoT communication protocols (e.g., MQTT, CoAP), and network security can contribute in this area.



Your Roles in an IoT Project *Cont`d...*

Full-Stack Developer:

Role: Develop end-to-end IoT applications, including user interfaces and backend services.

Students with skills in front-end technologies (e.g., HTML, CSS, JavaScript frameworks like React or Angular) and back-end development can play this role.



Your Roles in an IoT Project *Cont`d...*

Security Specialist:

Role: Ensure the security of IoT devices, data, and communication channels.

Students with a focus on cybersecurity, cryptography, and ethical hacking can contribute to making IoT systems secure.



Your Roles in an IoT Project *Cont`d...*

User Experience (UX) Designer:

Role: Design user interfaces for IoT applications.

Students with a background in design, HCI (Human-Computer Interaction), and front-end development can contribute in this area.



Your Roles in an IoT Project *Cont`d...*

IoT Architect:

Role: Design the overall architecture of IoT systems.

This role is suitable for students who have a broad understanding of IoT technologies and are interested in high-level system design.



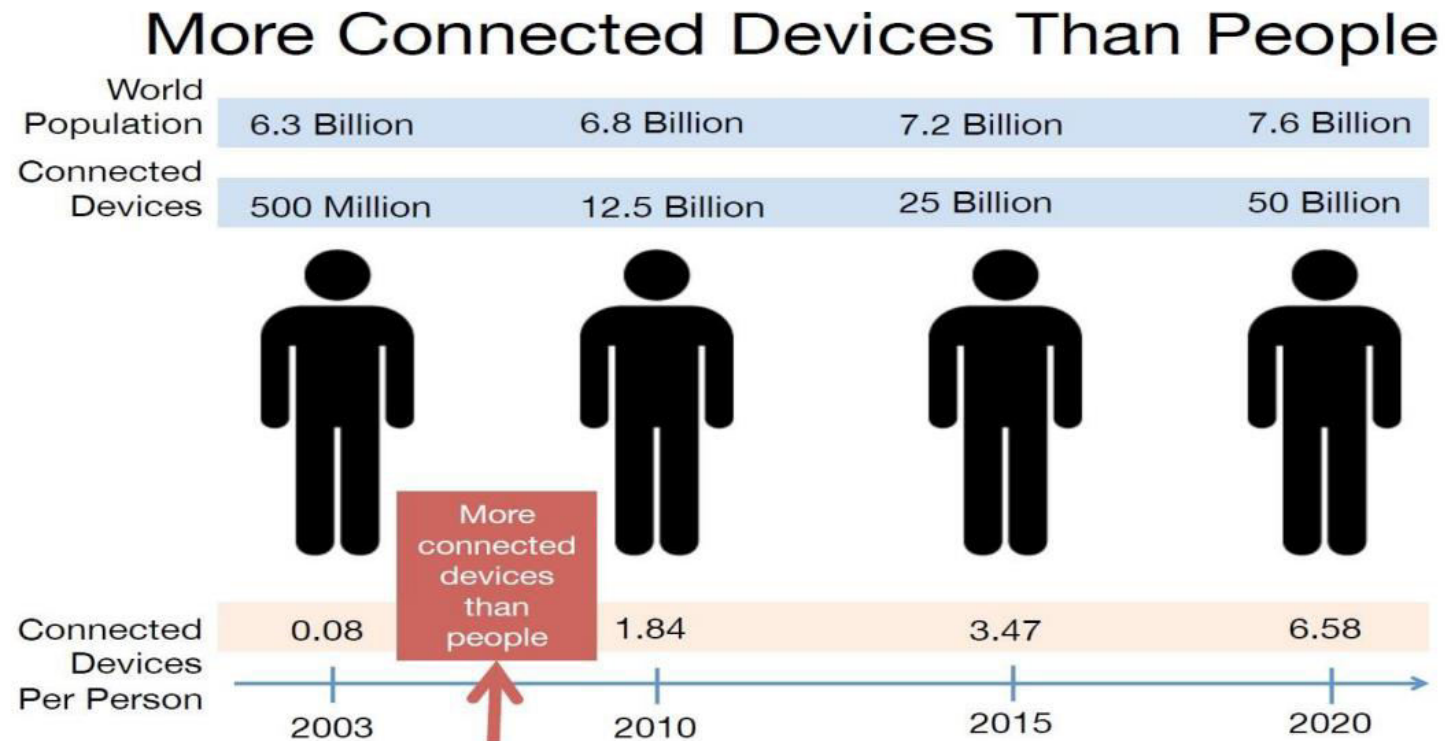
History

- IoT involves the addition of senses to computers.
- In the 20th century, computers were brains without senses.
- In the 21st century, computers are sensing things for themselves.



Genesis of IoT

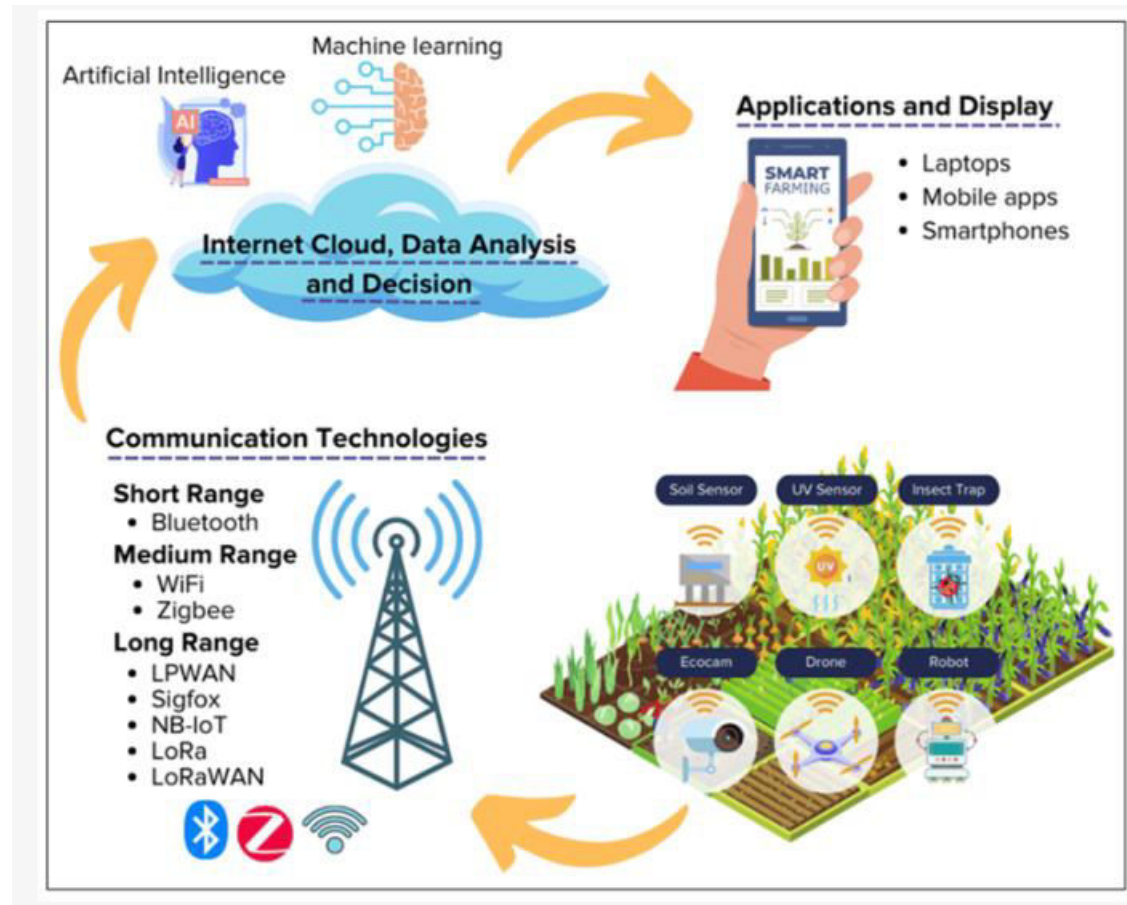
The age of IoT is started in 2008 and 2009. In these years, more things connected to the Internet than people in the world.



[Source: Cisco IBSG, April 2011]

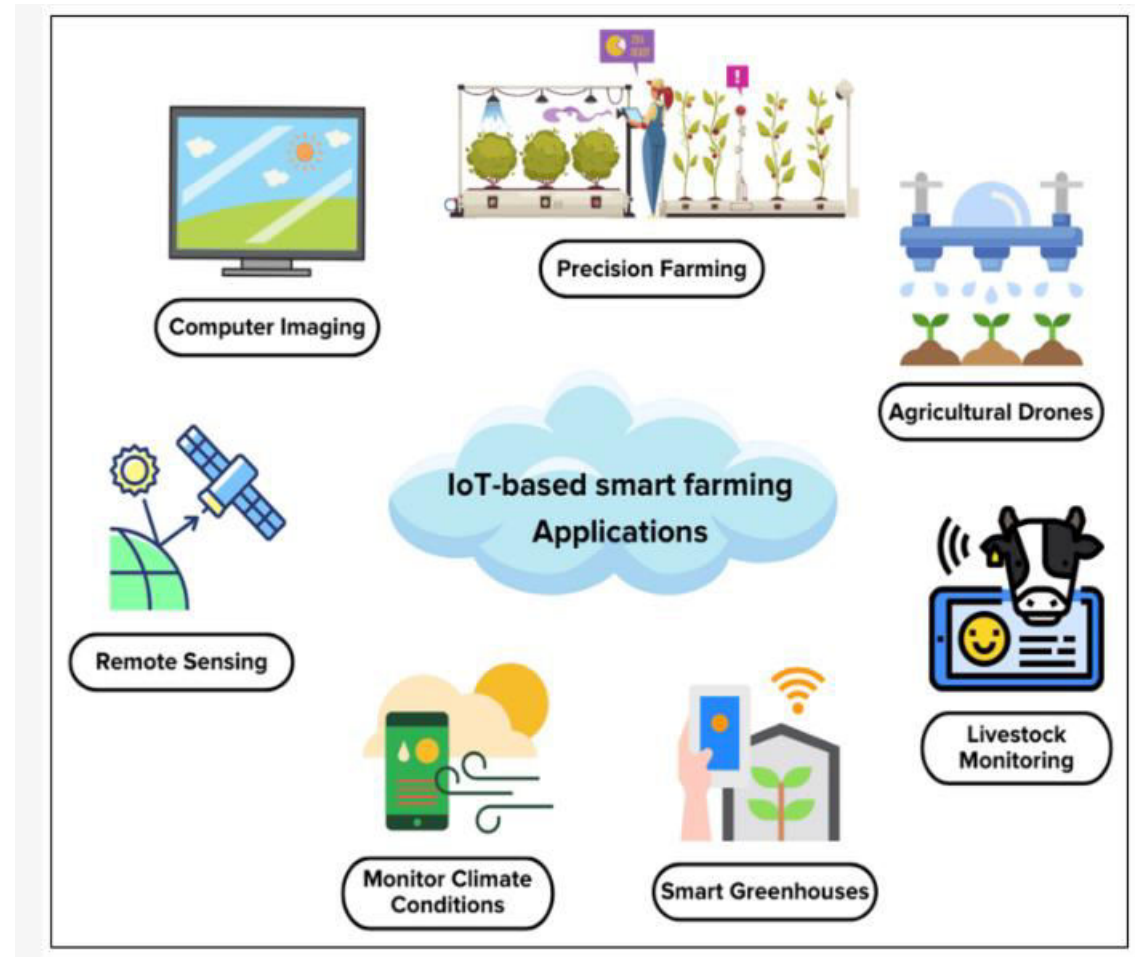
IoT Applications

IoT based smart farming



IoT Applications

IoT based smart farming



IoT Applications

IoT based smart classroom



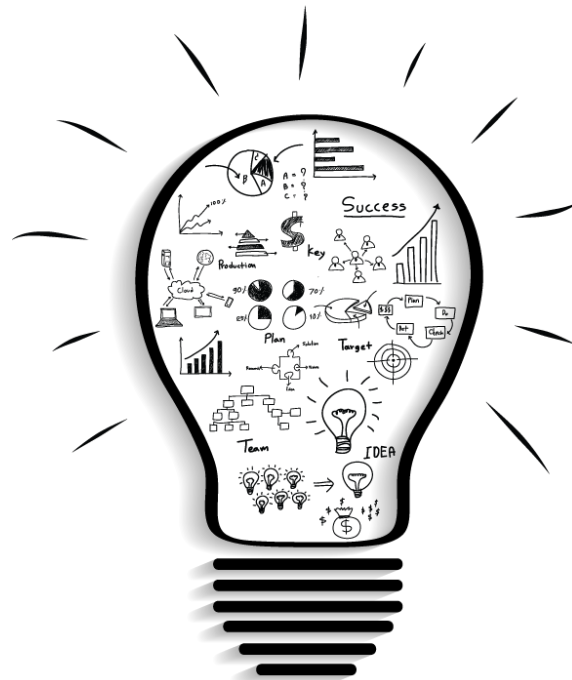
IoT Applications

Smart Campus

- 1. Environmental Control**
- 2. Automated Attendance System**
- 3. Smart Parking Management**
- 4. Smart Lockers**
- 5. Smart IoT Lab**
- 6. AR Equipped Systems**
- 7. Security of the Premises**

Your Project Ideas

Your project idea does not need to be entirely new. It can be an improvement to an existing solution. Focus on enhancing current technologies, optimizing processes, or integrating additional features that address specific challenges or inefficiencies.



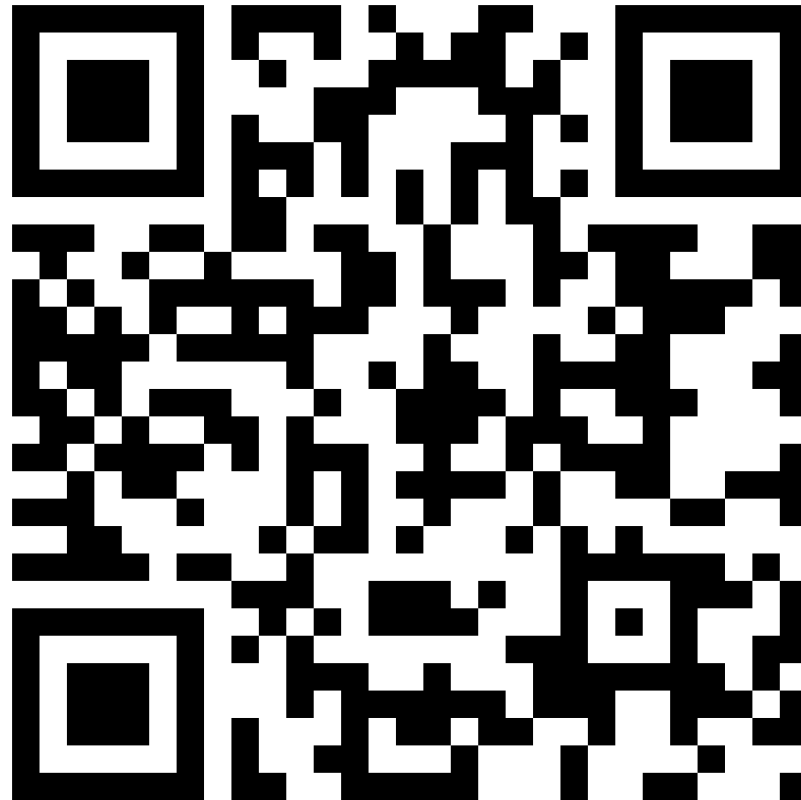
Your Project Ideas

Group A



Your Project Ideas

Group B



Questions ?



Thank you !

